A PROJECT REPORT
ON

"A PROSPECTIVE OBSERVATIONAL STUDY ON RISK ASSESSMENT AND MOTHER'S KNOWLEDGE ON NEONATAL JAUNDICE"

DISSERTATION

Submitted to

Jawaharlal Nehru Technological University, Hyderabad

In partial fulfillment of the requirement for the award of Degree of

DOCTOR OF PHARMACY

BY

GARGI MONDAL	(17BU1T0007)
KUNCHE SANDHYA	(17BU1T0010)
MANCHALA RAVI TEJA	(17BU1T0016)
RAJANALA RATHNAKAR	(17BU1T0021)

UNDER THE ESTEEMED GUIDANCE OF

HOSPITAL GUIDE

Dr. UDAY KUMAR KANDURI
MBBS, DCH, DNB,
MRCPCH(UK),
Department of pediatrics,
ESI Hospital Sanathnagar, Nacharam,
Hyderabad, Telangana State – 500076.

INSTITUTIONAL GUIDE

Dr. GABRIELA KEERTHANA GONDHI,
Pharm.D, MBA-HHSM, Assistant Professor
Department of Pharmacy Practice,
Marri Laxman Reddy Institute of Pharmacy,
Dundigal, Hyderabad,
Telangana State – 500043.

DEPARTMENT OF PHARMACY PRACTICE

MARRI LAXMAN REDDY GROUP OF INSTITUTIONS

MARRI LAXMAN REDDY INSTITUTE OF PHARMACY

(Approved by AICTE & PCI, New Delhi and Affiliated to JNTUH)

Dundigal - Gandimaisamma (V) & (M), Medchal (Dt), Hyderabad, Telangana - 500 043.

CERTIFICATE

This is to certify that the project work entitled "A PROSPECTIVE OBSERVATIONAL STUDY ON RISK ASSESSMENT AND MOTHER'S KNOWLEDGE ON NEONATAL JAUNDICE" is a bonafide work done by GARGI MONDAL (17BU1T0007), KUNCHE SANDHYA (17BU1T0010), MANCHALA RAVI TEJA (17BU1T0016), RAJANALA RATHNAKAR (17BU1T0021) in partial fulfillment of the requirements for the award of the degree of DOCTOR OF PHARMACY (PHARM.D) carried out in Department of Pediatrics, ESI Hospital – Sanathnagar (located at Nacharam) and in Department of Pharmacy Practice, Marri Laxman Reddy Institute of Pharmacy, Dundigal, Hyderabad.

Dr. N. JYOTHI, M. Pharm, Ph. D,

Professor & Principal,

Marri Laxman Reddy Institute of Pharmacy,

Dundigal, Hyderabad

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HOSPITAL GUIDE

Dr. UDAY KUMAR KANDURI
MBBS, DCH, DMB,
MRCPCH(UK),
Department of pediatrics,
ESI Hospital Sanathnagar, Nacharam,
Hyderabad, Telangana State – 500076.

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Dr. GABRIELA KEERTHANA GONDHI,
Pharm.D, MBA-HHSM, Assistant Professor
Department of Pharmacy Practice,
Marri Laxman Reddy Institute of Pharmacy,
Dundigal, Hyderabad,
Telangana State – 500043.

MARRI LAXMAN REDDY GROUP OF INSTITUTIONS

DECLARATION

We declare that this thesis entitled "A PROSPECTIVE OBSERVATIONAL STUDY ON RISK ASSESSMENT AND MOTHER'S KNOWLEDGE ON NEONATAL JAUNDICE" is an original report of my research, has been carried out by us in Department of Pediatrics, ESI Hospital–Sanathnagar (located at Nacharam) and in Department of Pharmacy Practice, Marri Laxman Reddy Institute of Pharmacy, Dundigal, Hyderabad for the award of the degree of DOCTOR OF PHARMACY (PHARM.D) has not been submitted for any previous degree. The experimental work is entirely our work; the collaborative contributions have been indicated clearly and acknowledged. Due references have been provided on all supporting literature and resources.

GARGI MONDAL (17BU1T0007)

KUNCHE SANDHYA (17BU1T0010)

MANCHALA RAVI TEJA (17BU1T0016)

RAJANALA RATHNAKAR (17BU1T0021)

Date:

Place: Hyderabad.

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We would like to thank our **friends** for their timely help and priceless support provided at every step during the project and throughout the course.

Further, we would like to thank all the patients for their trust, patience, and abidance. They have played a pivotal role in the study and their compliance with the study is most esteemed and commendable.

With regards,

Gargi Mondal (17BU1T0007)

Kunche Sandhya (17BU1T0010)

Manchala Ravi Teja (17BU1T0016)

Rajanala Rathnakar (17BU1T0021)

We dedicate our dissertation work wholeheartedly to our family and friends. A special feeling of gratitude to our parents for their unconditional love, support, and encouragement throughout our lifetime.

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ACRONYM	FULL FORM
RBC	Red blood cells
IgG	Immunoglobulin G
G6PD	Glucose-6-phosphate dehydrogenase
UV	Ultra violet
TSB	Total serum bilirubin

INTRODUCTION

1.INTRODUCTION

Neonatal jaundice is yellowish discoloration of skin, conjunctiva and sclera due to elevated serum or plasma bilirubin in the newborn period. It is typically a transient and mild event [1]. It is a condition where the total serum bilirubin increases above the 95th percentile during the 1st week after birth. 8% to 11% infants develop neonatal jaundice and 60% to 70% of healthy infants are seen with idiopathic symptoms. Elevated bilirubin levels are toxic for development of Central nervous system and causes neurological and behavioral impairment. It can be due certain factors such as gestational age, premature rupture of membranes, birth weight and maternal infections [2].

Figure 1 Newborn with jaundice [29]

ETIOLOGY

Neonatal jaundice is differentiated into several types.

Physiological jaundice occurs due to differences in bilirubin metabolism as there is increased production of bilirubin due to decreased lifespan of red blood cells, decreased bilirubin clearance due to deficiency of uridine diphosphate glucuronosyltransferase (UGT) and increased enterohepatic circulation.[1] It is observed from day 2 to 5 and disappears by 10-14days. [2]

Pathological jaundice observed on the first day of birth. There is a rapid increase in the bilirubin which can be due to immune mediated hemolysis as ABO and Rhesus incompatibility, non-immune mediated such as cephalohematoma, RBC membrane defects.[1] The increase in the serum bilirubin levels is higher than 5mg/dl/day and presence of this for greater than 14 days is considered as pathological jaundice.[2]

Breastfeeding babies when compared to artificial feeding babies, breast feeding babies have different patterns for jaundice. In the breastfed babies mild jaundice occurs at 24-72hrs of life, can take 10-14 days and may also reoccur in breastfeeding period. In 1/3rd of breastfed babies, clinical jaundice is seen in the third week after birth and can continue for 2 to 3 months in some. Cerebral lesions are caused as a large quantity of bilirubin accumulates in blood which is called nuclear jaundice. This may lead to certain conditions such as behavioral disorders, hearing loss and mental disorders. Decreased breastfeeding can increase severity physiological jaundice hence mothers are encouraged to breastfeed for 10-12 times a day. The breastfeeding in not recommended to stop until the levels increase more than 20mg/dl.[2]

Rhesus hemolytic disease is caused from maternal red cell alloimmunization when the baby born is Rh-positive to a Rh-negative mother and Rh-positive father. Here the mother's immunoglobulin antibodies i.e., IgG crosses the placenta and enters the fetal circulation causing several symptoms of mild to severe hemolytic anemia.[2]

ABO incompatibility is observed when mothers blood group is O and the newborn is of A or B blood group. It is observed in 15-20% of pregnancies and the jaundice is seen after 24hrs of birth.[2]

Infants with family history of G6PD deficiency has greater chances of severe jaundice . It is the deficiency of an enzyme in RBCs which plays a vital role in hexose monophosphate pathway.[2]

SYMPTOMS

- Yellowing in the sclera, soles of feet, palms and inside of their mouth.[2]
- The neonate will be sleepy.[2]
- The newborn may also not will to feed.[2]
- Newborn have dark, yellow urine and pale stools.[4]

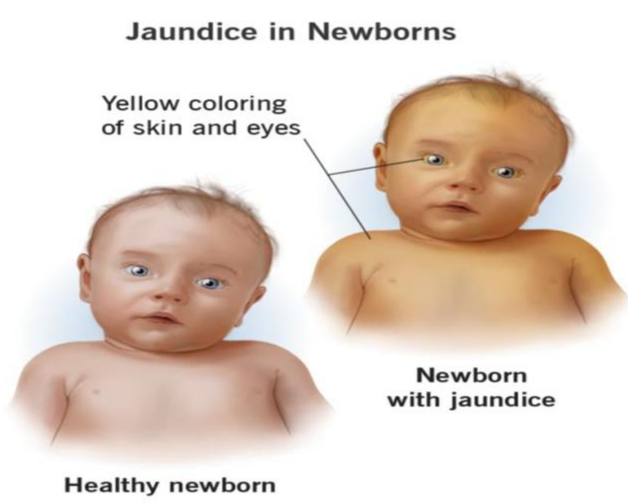


Figure 2 Symptoms of jaundice [30]

DIAGNOSIS

Visual examination: The diagnosis of neonatal jaundice varies during the 1st weeks of birth because visually all neonates have decrease in conjugation and excretion rate.[3]

Clinically diagnosing neonatal jaundice is difficult in dark skin color babies. The visual examination should be done including sclerae, blanched skin of limbs, torso, forehead and gums within 48 hours of birth under bright or natural light.[4] The staining of skin takes place in cephalo caudal direction. By using a digital pressure, the doctor should pale the skin and observe the subcutaneous tissue and underlying color of skin. If the neonate is observed to have yellow skin beyond thighs, then immediately the bilirubin levels should be checked.[2]

Measuring direct bilirubin: The measurement of conjugated bilirubin is very important in neonatal jaundice. Total and direct bilirubin levels are to be tested in every term baby who has jaundice for 14 days. The level of jaundice is based on values of direct bilirubin >35 μmol/L or more than 15% of total serum bilirubin.[4] Bilimeter is found useful in neonates as it uses spectrophotometry and identifies the total bilirubin levels.[4]

Stool examination: The stool color is to be checked for the neonates along with the bilirubin levels. Pale color stools are observed if the neonate is having jaundice.[4]

TREATMENT

Treatment options for jaundice in neonates include phototherapy, exchange transfusion and pharmacological treatment.[2]

Phototherapy: The efficacy of phototherapy depends on surface area exposed to phototherapy.

Double surface phototherapy may be more effective than single surface phototherapy.[2] Blue green fluorescent light is more effective than blue fluorescent light at reducing the phototherapy requirement in low birth weight babies after 24 hours with jaundice in first four days of life.[8]

Figure 3 Phototherapy [31]

Mechanism of photo therapy: The goal of phototherapy is to reduce the concentration of bilirubin. This is achieved by phototherapy to change the shape and structure of bilirubin and converting them to the molecules that can be easily excreted even there is abnormal conjugation. Absorption of light from skin and subcutaneous bilirubin induces a fraction of pigment that undergo many photochemical reactions that produce a low stereoisomer of bilirubin. These products formed are less lipophilic than bilirubin and can be excreted without conjugation in bile or urine.[10]

Photo isomerization occurs fast during the phototherapy and long before the level of plasma bilirubin declines, the isomers appear in blood.[10]

The common misconception is that UV light is used in phototherapy.[10]

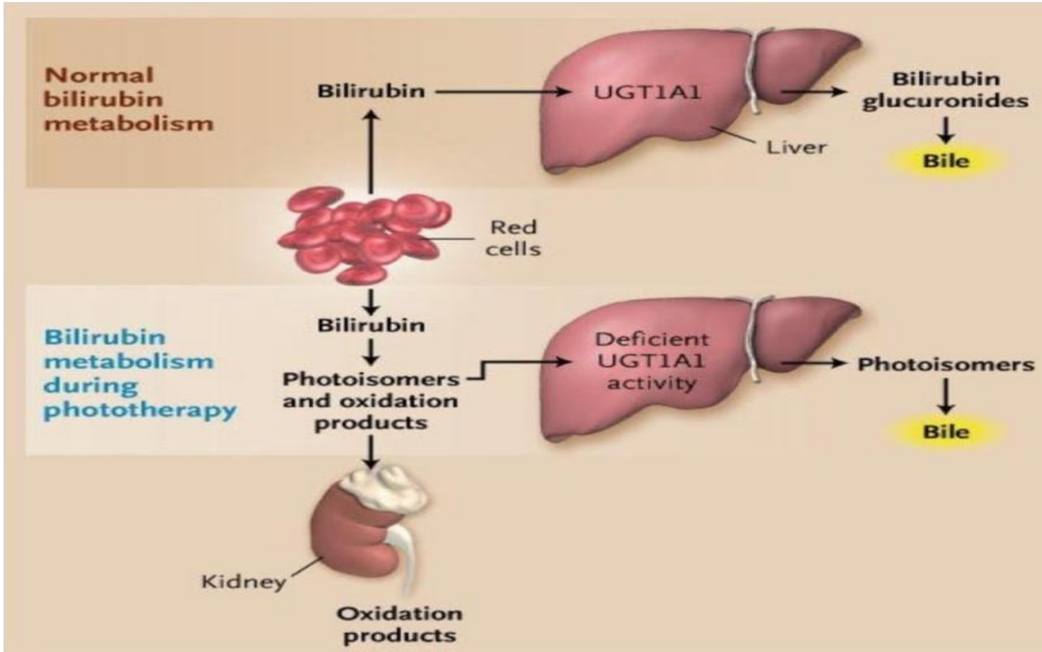


Figure 4 Mechanism of phototherapy [32]

Exchange transfusion: It could be done in infants with TSB levels of 25 mg/dl or greater and with acute bilirubin encephalopathy signs.[9] Bilirubin and hemolytic antibodies are removed by exchange transfusion. It includes Rh isoimmunization and ABO incompatibility.[2]

Figure 5 Exchange transfusion in neonate [33]

Pharmacological treatment:

a) Phenobarbitone: Processing of bilirubin includes hepatic uptake, conjugation and its excretion which is ameliorated by phenobarbitone, thus bilirubin decreases.[2]

   Dose: 5 mg/kg for 3-5 days [2]

b) Intravenous Immunoglobulin (IVIG): High dose IVIG (0.5 - 1 gm/kg) will decrease the needs of phototherapy and exchange transfusion in babies with Rh hemolytic disease.[2]

c) Metalloporphyrin: These are still experimental compounds but show promising results in hemolytic and non-hemolytic jaundice without notable side effects.[2]

COMPLICATIONS:

Complications are rare but are serious which include

- Cerebral Palsy:

  It is a disorder with posture, abnormal tone and movement of the body. Premature birth and low birth weight are the risk factors for cerebral palsy. [5]

- Kernicterus:

  It is the bilirubin induced brain damage in new born. The hippocampus, basal ganglia, geniculate bodies and cranial nerve nuclei are mostly affected. The risk of kernicterus increases with bilirubin levels >25mg/dl. When levels are > 30mg/dl leads to high risk and irreversible damage. [6]

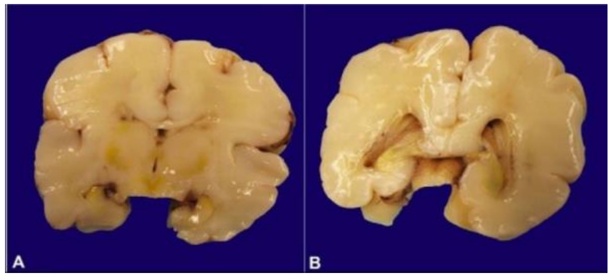


Figure 6 kernicterus: (A)yellow discoloration in floor of ventricles (B) lateral ventricles demonstrating minute hemorrhages [34]

- Deafness

PREVENTION:

- ✓ Breastfeeding should be encouraged by mother to baby frequently at least 8 to 12 times a day.[4]
- ✓ Programmes should be conducted at health care institutions about the jaundice in neonates and breastfeeding support for mothers.[7]

LITERATURE

REVIEW

2.LITERATURE REVIEW

2.1 ***Tavakolizadeh R, et al. have performed a cross-sectional study in Ziaeian and Imam Khomeini Hospitals, Tehran titled, "Maternal risk factors for neonatal jaundice: A hospital based cross-sectional study in Tehran'' published in European journal of translational myology, 2018, which states that*** Neonatal Jaundice happens after birth which is common and unpreventable. The study intended to assess the maternal risk factors that are contributing to increased bilirubin. The study took 2207 term newborns who were less than 15 days of age and had bilirubin levels more than 15 mg/dl. The results showed that maternal age, BMI, weight, WBC, PLT, Hb, numbers of pregnancies, prolonged delivery and birth in first pregnancy were observed to be correlated significantly with hyperbilirubinemia.[10]

2.2 ***Murekatete C, et al. have performed retrospective cross-sectional design in district hospital in Rwanda titled "Neonatal Jaundice risk factors at a District Hospital in Rwanda'' published in Rwanda Journal of Medicine and Health Sciences, 2020, which states that*** Hyperbilirubinemia can cause life-long brain damage so screening and prompt treatment is necessary. The goal of the study was to check neonatal jaundice risk factors in newborns. 210 files were selected by stratified proportional sampling for the years 2016-2018. The study discovered that Gestational age, blood group incompatibilities, Birth weight, neonatal gender, infections, prematurity and C-section were associated with neonatal jaundice.[11]

2.3 ***Goodman, et al. have performed a descriptive cross - sectional study in Mosan - Okunola community, Lagos, Nigeria titled, "Neonatal Jaundice: knowledge, attitude and practices of mothers in Mosan - Okunola community, Lagos, Nigeria" published in Nigerian postgraduate medical journal, 2015, which states that*** there were many misconceptions about neonatal jaundice where knowledge gaps need to be addressed and filled. The study intended to judge the knowledge and preventive practices of mothers in the Mosan - Okunola community. This study included 358 women and found that the awareness of neonatal jaundice was 75.4%. If the baby developed neonatal jaundice 90.4% mothers had willingness to take the baby to hospital. The study also found out that the awareness was high but the knowledge about causes, complications and danger signs was low.[12]

2.4 ***Boo NY, et al. have performed a cross-sectional study in Tuanku Jaafar Hospital, titled "Malaysian mothers knowledge & practices on care of neonatal jaundice'' published in Med J Malaysia, 2011, which states that*** excessive rise of unconjugated bilirubin which is neurotoxic can cause cerebral palsy, hearing loss and death in newborns. The focus of the study was to estimate the gaps in knowledge and care of neonatal jaundice among the Malaysian mothers including races like Malays, Chinese and Indians. 400 participants (200 Malay, 100 Chinese and 100 Indians) were taken in the study. The study discovered that Malay mothers are more knowledgeable on neonatal jaundice than Chinese or Indian mothers. It also showed that there is a big gap of knowledge on neonatal jaundice and a need to improve the education of mothers on neonatal jaundice.[13]

2.5 ***Asmamaw Demis, et al. have performed a cross-sectional study in Amhara region, Ethiopia, titled "Knowledge on neonatal jaundice and its associated factors among mothers in north Ethiopia: a facility-based cross-sectional study" published in BMJ open, 2021, which states that*** poor decision and slow seeking of medical care by mothers can be due to lack of information or wrong information. The intention of the study was to determine mother's knowledge and associated factors on neonatal jaundice in Northern Ethiopia. The study included 380 mothers using systematic random sampling techniques. It found out the overall mother's knowledge was 39.2%. The mothers who had a favorable approach towards neonatal jaundice have good knowledge. mother's attitude on Neonatal jaundice, Antenatal care follow up, child history of Neonatal jaundice, current child history of Neonatal jaundice, and residence were variables that had a significant association with mother's knowledge on Neonatal jaundice. The study inferred that awareness about prevention can be increased by conducting education programs on neonatal jaundice.[14]

2.6 ***Taco J prins, et al. have performed a cross-sectional study at Shoklo malaria research unit on Thailand- Myanmar border, titled "A survey of practice and knowledge of refugee and migrant pregnant mothers surrounding neonatal jaundice on Thailand - Myanmar border" published in Journal of tropical pediatrics, 2017, which states that*** G6PD deficiency and hemolytic triggers are causes of severe neonatal jaundice which can be prevented by increasing awareness among the mothers. The study wanted to determine the use of mothballs among mothers in a population. 522 pregnant women were taken in the study. 40% of the population used mothballs to store baby clothes and blankets. The use of hemolytic agents in populations with G6PD deficiency needs to be decreased. It was found mothers with previous children who had neonatal jaundice were more knowledgeable. The mortality and morbidity due to neonatal jaundice can be decreased by identifying and increasing awareness about the preventable risk factors like hemolytic agents.[15]

2.7 ***Hassan Boskabadi, et al. have performed a systematic review, titled "Evaluation of maternal risk factors in neonatal hyperbilirubinemia" published in archives of Iranian medicine,2020, which states that*** controlling jaundice can be effective if the predisposing factors for neonatal jaundice are identified. The study systematically reviewed maternal risk factors in neonatal jaundice. This study took 17 related articles from different databases. The study concluded with list of maternal risk factors includes pregnancy problems(prematurity, premature rupture of membrane, negative maternal Rh), maternal diseases(hypertension, diabetes, preeclampsia, gestational infections), labor problems(type of delivery, cephalohematoma, place of delivery, delayed labor, delivery trauma, complications), breastfeeding problems(breast problems, reduced feeding, inappropriate breastfeeding technique) and other factors like maternal age, first delivery, abortion history.[16]

2.8 ***Assefa GG, et al. have performed a case control study in public general hospitals of central zone, Tigray and northern Ethiopia titled, "Determinants of neonatal jaundice among neonates admitted to neonatal intensive care unit in public general hospitals of central zone, Tigray, northern Ethiopia, 2019: a case control study" published in biomed research international, 2020, which states that*** severe hyperbilirubinemia can lead to bilirubin encephalopathy or kernicterus which increases risk of mortality and neurological damage like cerebral palsy, intellectual difficulties, hearing loss or gross developmental delays. The goal of the study was to identify the determinants of neonatal jaundice in NICU. The study took 272 medical records of jaundiced neonates (91) and control as neonates without jaundice (181). The study found out Rh incompatibility, obstetric complication, asphyxia, low birth weight, polycythemia and breastfeeding were the determinants of neonatal jaundice. The study concluded that to prevent long term complications and death in neonates early identification, prevention and treatment are necessary.[17]

2.9 ***Brits H, et al. have performed a cross-sectional study in national district hospital in Bloemfontein, titled, "The prevalence of neonatal jaundice and risk factors in healthy term neonates at national district hospital in Bloemfontein'' published in African journal of primary health care and family medicine, 2018, which states that*** neonatal jaundice is self-limiting but increased levels of bilirubin can cause permanent neurological damage. The study wanted to determine the risk factors and prevalence of neonatal jaundice in term neonates. The study included 96 mother-neonate pairs born at National district hospital in Bloemfontein. The study inferred that neonatal jaundice prevalence was 55.2%. The risk factors which were identified by the study were Asian race, induction of labor with oxytocin, instrumental delivery, infant bruising and exclusive breastfeeding.[18]

2.10 ***Egube BA, et al. have performed a descriptive cross-sectional study in ante-natal clinic of University of Benin teaching hospital in Benin city, Nigeria, titled "Neonatal***

jaundice and its management: knowledge, attitude and practice among expectant mothers attending antenatal clinic at university of Benin teaching hospital" Benin city, Nigeria published in Nigerian journal of clinical practice, 2013, which states that one of the major causes for neonatal mortality is neonatal jaundice. Neonatal jaundice very high in sub-Saharan Africa, Asia and Latin America. The study included 389 expectant mothers attended ante-natal clinic in 2005.The study learned that 85.9% knew about neonatal jaundice, 77.4% knew how to detect neonatal jaundice. 71.7% knew of treatment of neonatal jaundice, 67% knew the complication. The study stated that knowledge level about Neonatal jaundice was influenced by the education level and previous babies with Neonatal jaundice.[19]

2.11 ***Su Yurn Ug, et al. have performed a cross-sectional observational study in hospital Teluk Intan, Malaysia, titled "what do mothers know about neonatal jaundice? knowledge attitude and practice of mothers in Malaysia" published in The Medical journal of Malaysia, 2014, states that*** kernicterus is reemerging which has at least 10% mortality and 70% morbidity. The study included 198 mothers out of 238 eligible mothers due to variable reasons to be excluded. The study discovered that 80.3% mothers relied on health nurses to detect jaundice, only 8.1% of mothers were able to recognize Neonatal jaundice. 22.2% of mothers were able to find six complications of Neonatal jaundice (convulsion, kernicterus, deafness, mental retardation, death and physical handicap). The study inferred that there are a lot of misconceptions of Neonatal jaundice, so it is necessary to educate mothers about Neonatal jaundice. [20]

2.12 ***Sayed yosef mojtahedi, et.al, have performed a cross sectional study in Ziyaeian hospital and Imam Khomeini hospital, Iran, titled "Risk factors associated with neonatal jaundice a cross-sectional study from Iran" states that*** for proper management of Neonatal jaundice identifying the predisposing factors is very essential. The goal of the study was to establish the risk factors of jaundice in neonates. 200 mothers and neonates were taken into the study. The study learned that mother's hemoglobin, platelets, WBC, gestational age, G6PD deficiency, T4 and TSH levels were significantly linked with hyperbilirubinemia. So, it is very important to identify the predisposing factors which can help in early diagnosis and reduce complication.[21]

2.13 ***Kolou H.amegan-Aho, et al. have performed a cross-sectional study at two antenatal clinics in Accra, titled "Neonatal jaundice awareness perception and preventive practices in expectant mothers" published in Ghana medical journals, 2019, states that*** sepsis, G6PD deficiency, blood group incompatibility and cephalhematoma are common causes of unconjugated hyperbilirubinemia. The study intended to find out attitude, knowledge and practice of neonatal jaundice among expectant mothers. The study involved 175 expectant mothers carried out from 1st and 17th November 2013. In this study, 77.1% of subjects heard about Neonatal jaundice among them only 72.6% knew not less than one symptom of Neonatal jaundice, 52% of subjects knew about treatment of Neonatal jaundice, 63.5% knew complications and 49.6% knew danger signs. 92.6% did not know causes of Neonatal jaundice. The study deduced that the expectant mothers were aware of Neonatal jaundice but lacked knowledge regarding causes, signs, treatment and complications.[22]

2.14 ***Asmamaw Demis Bizureh, et al. have performed unmatched case control study design at referral hospitals in Amhara region, Northern Ethiopia,*** *titled* ***"Determines of neonates admitted to five referred hospitals in Amhara region Northern Ethiopia: An unmatched case control study" published in British medical journals pediatrics open, 2020, states that*** Neonatal jaundice is related with readmissions in first 7 days of life and significant mortality. The goal of the study was to recognise determinant factors of Neonatal jaundice in neonates. The study looks 447 neonates (149 as cases and 298 as control) from 1 march- 30 July 2019. The study found out prolonged labor, low birth weight, sepsis, birth asphyxia, hypothermia and sex were triggers of Neonatal jaundice. The study inferred those strategies should be formulated to identify high risk neonates which will aid in early recognition and control of Neonatal jaundice and reduce its incidence.[23]

2.15 ***Eriko shinohara, Yaeko katoaka have performed a retrospective cohort study at a birth center in Japan,*** *titled* ***"Prevalence and risk factors for hyperbilirubinemia among newborns from a low-risk birth setting using delayed cord clamping in Japan'' published in Japan journals of nursing science, 2020, states that*** delayed cord clamping increases risk of hyperbilirubinemia which require phototherapy. The aim of study was to estimate the risk factors/determinants of Neonatal jaundice. 1,211 patient records were taken from march 2006 to October 2014. The study found cephalohematoma, delay of meconium elimination, previous history of phototherapy of siblings and primiparity were the risk factors for Neonatal jaundice.[24]

2.16 ***Stephanee Campbell Wagemann, Patricia Mena Nannig performed a retrospective study in Dr. Sotero del Rio asistencial complex, Chile, titled "Severe hyperbilirubinemia in newborn, risk factors and neurological outcomes published in rev chil pediatr.,2019, states that*** bilirubin encephalopathy have decreased due to phototherapy and exchange transfusion. The study intended to find out the risk factors and incidence of Neonatal jaundice with bilirubin levels equal to or higher than 20 mg/dl. The study took 593 neonates with bilirubin levels equal or higher than 20 mg/dl between January 2013 and December 2016. The study found out excessive weight loss, prematurity, male sex and blood group incompatibility were the determinants for severe hyperbilirubinemia.[25]

2.17 ***Gitta arinda kebidana, et al. have performed a literature review, titled "Risk factors neonatal jaundice - literature review" published in nursing update,2021, states that*** the most common morbidity in neonates in Neonatal jaundice. The goal of the study was to review all the literature related to risk factors of jaundice in neonates. After screening of the literature, they included 14 articles. The neonatal jaundice risk factors were G6PD deficiency, ABO incompatibilities, obese mother, gestational age and breastfeeding. The study stated that general screening of neonates with TSB or skin bilirubin levels before discharge and good breastfeeding techniques can reduce the risk of Neonatal jaundice.[26]

2.18 ***Olusanya bo, et al. have performed a systematic review and meta-analysis titled "Risk factors for severe neonatal hyperbilirubinemia in low and middle-income countries: A systematic review and meta-analysis" published in PLOS ONE, 2015, states that*** compared to high income countries low-income countries bear most burden of severe

Neonatal jaundice which can be characterized by mortality, morbidity and neurodevelopmental disorders. The aim of study was to identify the risk factors which contribute to the burden of severe hyperbilirubinemia in low- and middle-income countries. They included 13 studies from January 1990 to June 2014. The study found out the maternal risk factors such as place of delivery, primiparity, ABO incompatibility and Rh incompatibility. The neonatal factors associated with hyperbilirubinemia were UGT1A1 polymorphism, gestational age, G6PD deficiency, weight on admission, TCB/TSB levels and sepsis. The study reported that if the risk factors are identified, Neonatal jaundice can be treated effectively.[27]

2.19 ***May-jen huang, et al. have performed a case control study in Taiwan, titled "Risk factors for severe hyperbilirubinemia in neonates" published in Pediatric research, 2004, states that*** Asians have more levels of plasma conjugated bilirubin than whites and black population. The study took 72 neonates and 100 controlled subjects. The study found out ABO incompatibility, premature birth, infections, cephalohematoma, G6PD deficiency, asphyxia, breast feeding and variant of UDP glucuronosyltransferase 1A1(UGT1A1 gene).[28]

AIMS AND OBJECTIVES

3.AIMS AND OBJECTIVES

AIM:

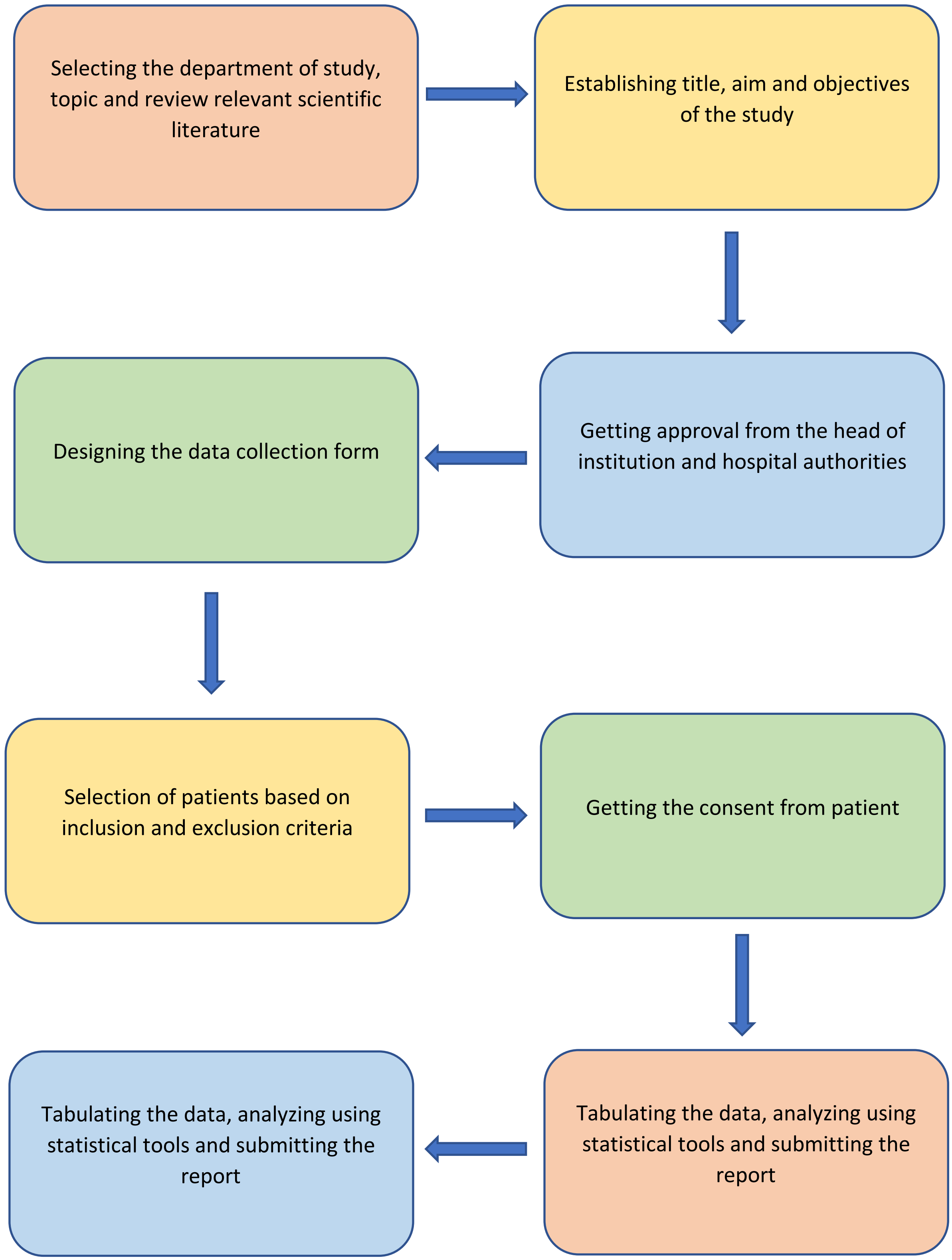
To assess the risk and mother's knowledge on neonatal jaundice.

OBJECTIVES:

- To assess the risk of jaundice in neonates.
- To assess mother's knowledge on neonatal jaundice.
- To compare full term with pre term neonates both having neonatal jaundice.
- To educate mothers regarding neonatal jaundice and increase their awareness.

PLAN OF WORK

4.PLAN OF WORK



METHODOLOGY

5.METHODOLOGY

5.1 STUDY DESIGN

This study design is prospective observational study.

5.2 STUDY SITE

The study will be conducted in IN PATIENT and OUT PATIENT facilities of Department of Pediatrics, ESI Hospital, a 500 bedded tertiary Care Hospital located in Nacharam, Hyderabad, Telangana.

5.3 STUDY PERIOD AND DURATION

The study period is of 6 months i.e., from November 2021 to April 2022.

5.4 STUDY POPULATION

106 patients visiting in-patient and out-patient facilities of Department of pediatrics, ESI hospital Sanathnagar located at Nacharam diagnosed with neonatal jaundice along with their mother were recruited in the study. Out of 106 subjects recruited in the study, 11 subjects we excluded due to incomplete data. Therefore, we completed the study with 95 subjects.

5.5 SELECTION OF SUBJECTS AND RECRUITMENT PROCEDURE

The study subjects will be selected from Department of Pediatrics based on the following inclusion and exclusion criteria.

5.6 INCLUSION CRITERIA:

- ✓ All the babies along with their mothers born during the study period.
- ✓ Babies of age <28 days.
- ✓ Mothers who can understand, answer the questions and willing to participate.

5.7 EXCLUSION CRITERIA:

- ✓ Babies of age > 28 days.
- ✓ Babies who are sick and ventilated.
- ✓ Mothers who are not willing to participate.
- ✓ Mothers who can't give complete information.

5.8 DROP OUTS:

Out of 106 subjects recruited in the study, 11 subjects we excluded due to incomplete data. Therefore, we completed the study with 95 subjects.

5.9 PATIENT DATA COLLECTION FORM:

A Suitable patient data collection form was prepared with the help of a physician. It includes patient demographic data (name, age, gender, weight, blood group, pre/full term, medical history) mother's demographic data (name, age, weight, blood group, lactation, personal history, medical and medication history, knowledge about jaundice).

5.10 STATISTICAL TOOLS:

We used Microsoft excel to perform chi-square test and t test. We calculated mean, standard deviation and p value using excel.

5.11 PATIENT COUNSELLING:

- ✓ All the mothers included in the study were given education about neonatal jaundice. The counselling session was 5-10 minutes. The mothers were counselled in a language they understand (Telugu, Hindi or English).
- ✓ They were counselled regarding causes, signs and symptoms, complications and prevention of neonatal jaundice.
- ✓ Patient information leaflet was given to the mothers at the end of the counselling session.

RESULTS

6.RESULTS

Out of 106 subjects (mother and neonate pair) recruited in the study, 11 subjects we excluded due to incomplete data. Therefore, we completed the study with 95 subjects.

6.1 DISTRIBUTION OF STUDY POPULATION(MOTHERS) BASED ON AGE

AGE	NUMBER OF SUBJECTS	PERCENTAGE
20-24	23	24.21%
25-29	43	45.26%
30-34	21	22.11%
35-39	8	8.42%
TOTAL	95	100.00%

Table 1 Distribution of study population(mothers) based on age

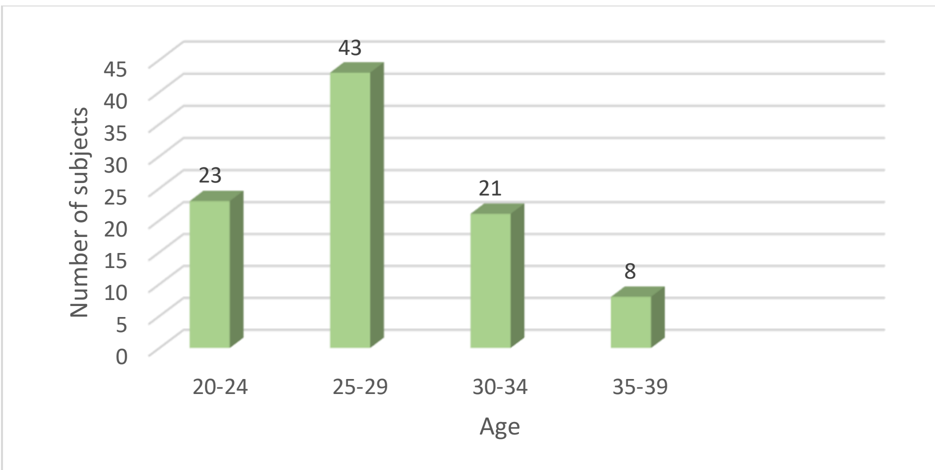


Figure 7 distribution of study population(mothers) based on age

The most numbers of study subjects were in 25-29 age group with 45.26% of the total study population. The least number of study subjects were in 35-39 age group with 8.42% of the total study population.24.21% of study population were of 20-24 age group and 22.11% were 30-34 age group.

6.2 DISTRIBUTION OF STUDY POPULATION(NEONATES) BASED ON SEX

SEX	NUMBER OF SUBJECTS	PERCENTAGE
MALE	49	51.58%
FEMALE	46	48.42%
TOTAL	95	100.00%

Table 2 Distribution of study population (neonates) based on sex

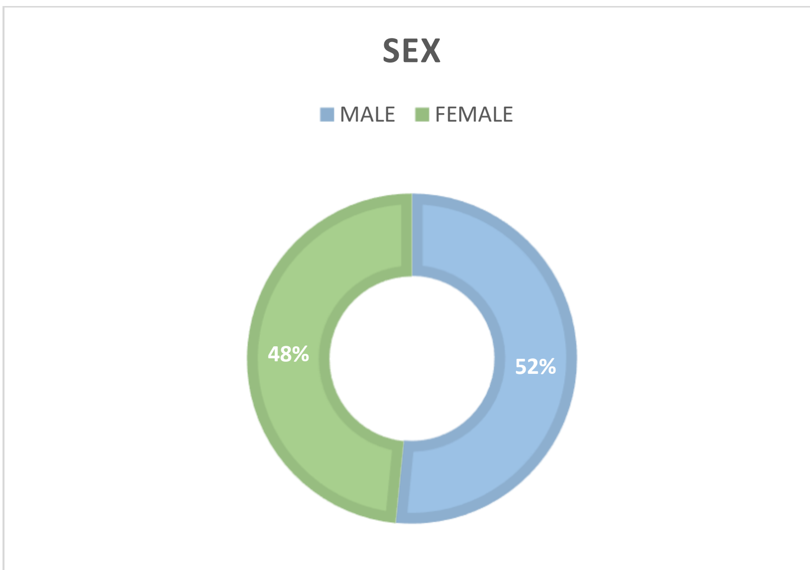


Figure 8 Distribution of study population(neonates) based on sex

In the total number of study subjects, 49(51.58%) were male neonates and 46(48.42%) were female neonates.

6.3 DISTRIBUTION OF STUDY POPULATION BASED ON BIRTH WEIGHT AND PRE-TERM OR FULL TERM

BIRTH WEIGHT	FULL TERM	PRE-TERM	TOTAL
2-2.2	-	3	3
2.2-2.4	6	3	9
2.4-2.6	8	4	12
2.6-2.8	7	5	12
2.8-3	6	7	13
3-3.2	22	4	26
3.2-3.4	6	5	11
3.4-3.6	6	1	7
3.8-4	2	-	2
GRAND TOTAL	63	32	95

Table 3 Distribution of study population based on birth weight and preterm or full term

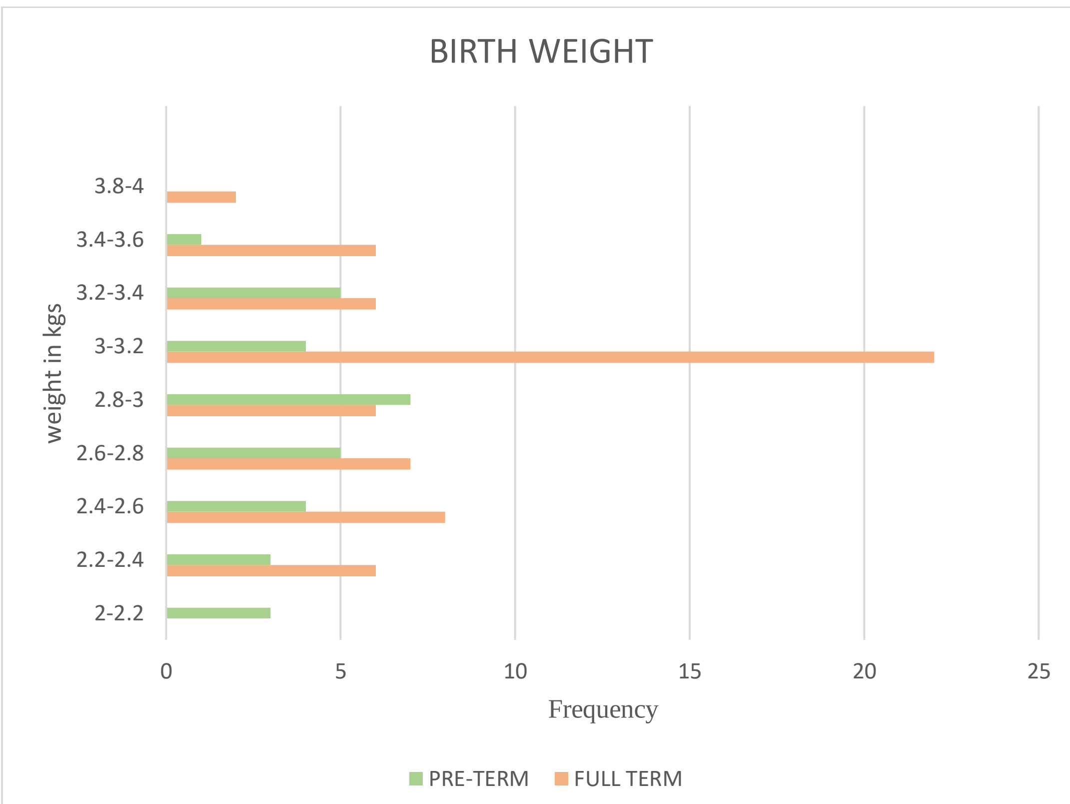


Figure 9 Distribution of study population based on birth weight and preterm or full term

In the total study population, the majority of full-term neonates had birth weight of 3-3.2 kgs and least number of study subjects were 2.2-2.4 kgs. In pre term neonates, majority of study subjects had birth weight of 2.8- 3 kgs and least number of them had birthweight of 2-2.2 kgs.

6.4 DISTRIBUTION OF STUDY POPULATION BASED ON RESIDENCY

RESIDENCY	FREQUENCY	PERCENTAGE
RURAL	16	16.84%
URBAN	79	83.16%
TOTAL	95	100.00%

Table 4 Distribution of study population based on residency

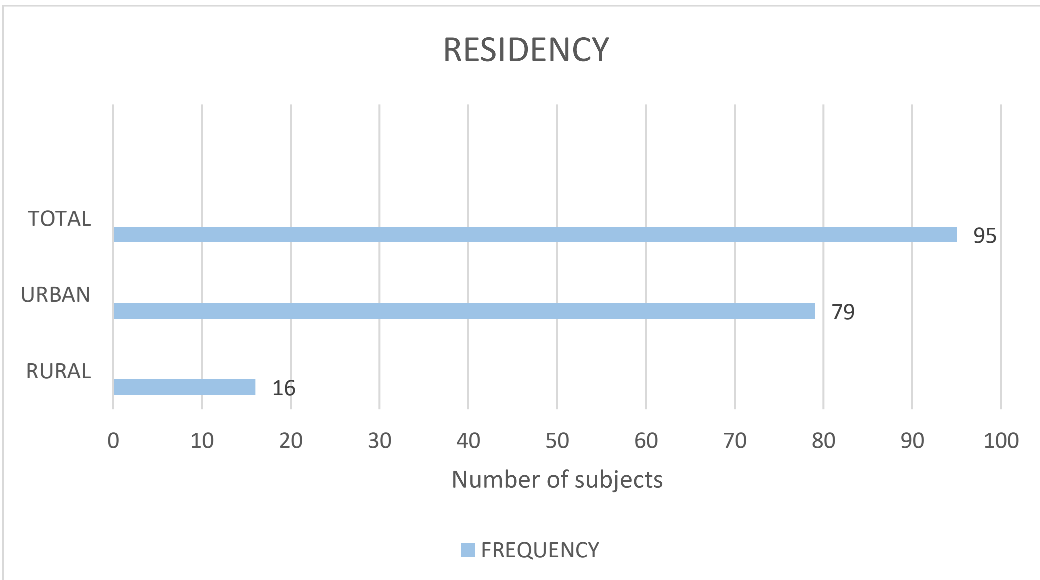


Figure 10 Distribution of study population based on residency

In the total population of subjects, majority of subjects were from urban area with 83.16% and 16.84% of subjects belonged to rural area.

6.5 DISTRIBUTION OF STUDY POPULATION BASED ON THEIR MARITAL LIFE

MARITAL LIFE	NUMBER OF SUBJECTS	PERCENTAGE
NON-CONSANGUINEOUS	82	86.32%
CONSANGUINEOUS	13	13.68%
TOTAL	95	100.00%

Table 5 Distribution of study population based on their marital life

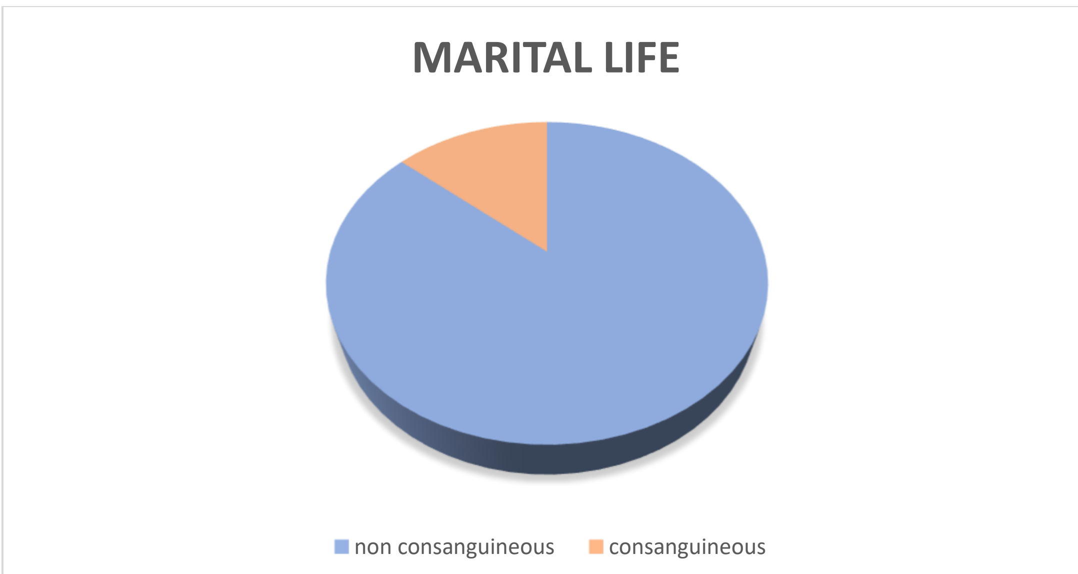


Figure 11 Distribution of study population based on their marital life

n the total number of the study subjects the marital life of 82(86.32%) subjects was of non-consanguineous and 13(13.68%) subjects of consanguineous.

6.7 DISTRIBUTION OF STUDY POPULATION BASED ON EDUCATION QUALIFICATION

EDUCATION QUALIFICATION	NUMBER OF PATIENTS	PERCENTAGES
POST GRADUATION	9	9.47%
GRADUATION	46	48.42%
DIPLOMA	2	2.11%
INTERMEDIATE	13	13.68%
SCHOOL	25	26.32%
TOTAL	95	100.00%

Table 7 Distribution of study population based on education qualification

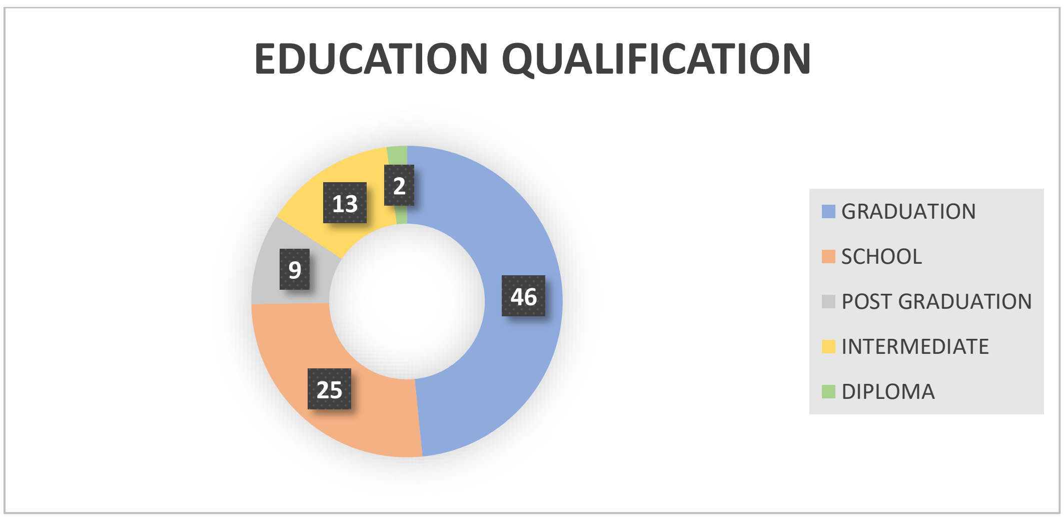


Figure 12 Distribution of study population based on education qualification

During the study period, it was found that 46 study subjects have completed their education till graduation, 9 subjects till post-graduation, 13 completed till intermediate, 2 of them completed their diploma and 25 subjects completed their schooling.

6.7 DISTRIBUTION OF STUDY POPULATION BASED ON TYPE OF DELIVERY

TYPE OF DELIVERY	NUMBER OF SUBJECTS	PERCENTAGE
NORMAL DELIVERY	6	6.32%
LOWER SEGMENT CESAREAN	89	93.68%
TOTAL	95	100.00%

Table 7 Distribution of study population based on type of delivery

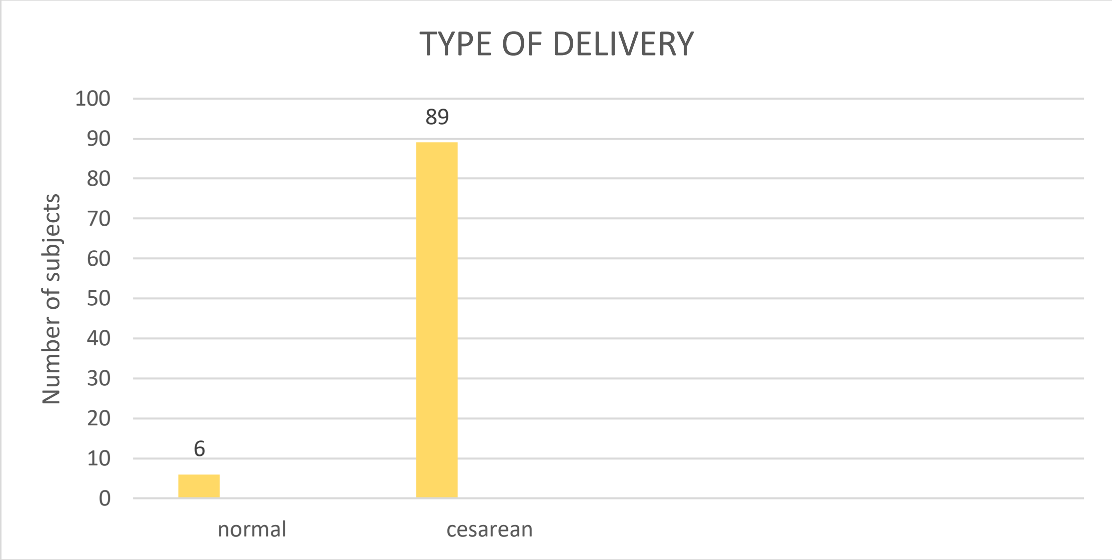


Figure 13 Distribution of study population based on type of delivery

The most number of study subjects had undergone cesarean type of delivery with 93.68% and the rest of the subjects had undergone normal delivery with 6.32%.

6.8 DISTRIBUTION OF STUDY POPULATION BASED ON INTERPREGANANCY GAP

INTERPREGNANCY GAP (YEARS)	NUMBER OF SUBJECTS	PERCENTAGE
0-1	34	35.79%
1-2	16	16.84%
2-3	14	14.74%
3-4	13	13.68%
4-5	8	8.42%
5-6	4	4.21%
6-7	4	4.21%
7-8	1	1.05%
8-9	1	1.05%

Table 8 Distribution of study population based on interpregnancy gap

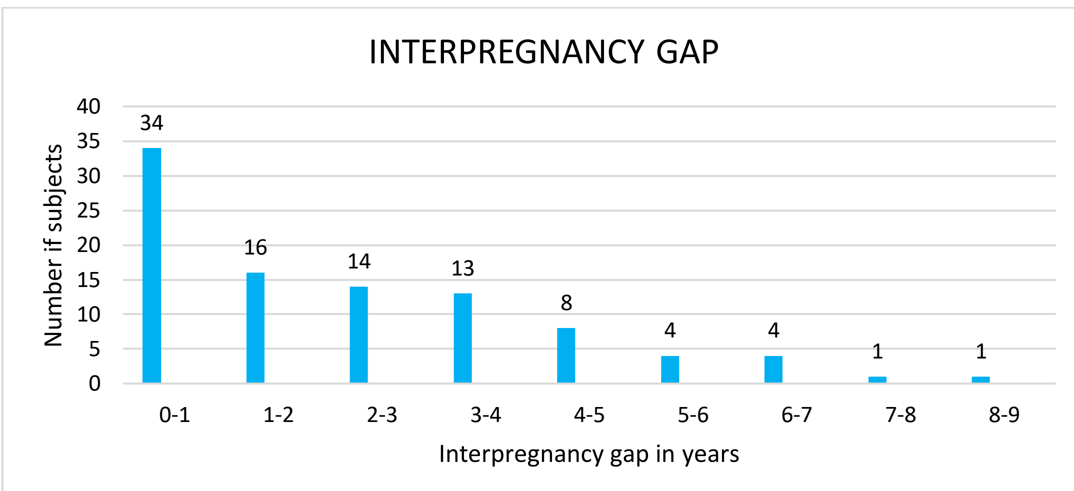


Figure 14 Distribution of study population based on interpregnancy gap

During the study, majority of the study subjects had an interpregnancy gap of 0-1 year with 35.79%, 16.34% of them with 1-2 years, 14.74% with 2-3 years, 13.68% with 3-4 years, 8.42% with 4-5 years, 4.21% with 5-7 years and lowest number of subjects had 7-9 years gap with 1.05%.

6.9 DISTRIBUTION OF STUDY POPULATION BASED ON DURATION OF JAUNDICE

DURATION OF JAUNDICE	NUMBER OF SUBJECTS	PERCENTAGE
1 DAY	21	22.11%
2 DAYS	39	41.05%
3 DAYS	24	25.26%
4 DAYS	10	10.53%
5 DAYS	1	1.05%
TOTAL	95	100.00%

Table 9 Distribution of study population based on duration of jaundice

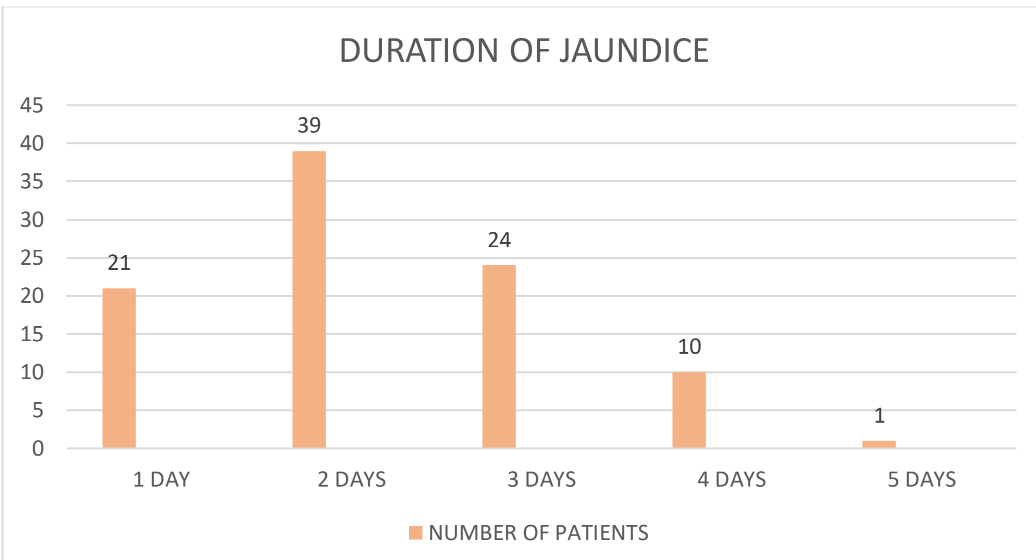


Figure 15 Distribution of study population based on duration of jaundice

In the study period, majority of the subjects had highest duration of jaundice for 2 days with a percentage of 41.05, 1.05% of the subjects had jaundice for 5 days, 10.53% of them had a duration of jaundice for 4 days and 25.26% of subjects had duration for 3 days.

6.10 DISTRIBUTION OF STUDY POPULATION BASED ON RISK FACTORS RELATED TO MOTHER

RISK FACTOR	NUMBER OF SUBJECTS	PERCENTAGE
NO FEEDING DURING 1ST HOUR AFTER DELIVERY	61	64.21%
NO PROPER LACTATION	16	16.84%
BREAST PROBLEMS	4	4.21%
PRE-MATURE RUPTURE OF MEMBRANE	11	11.58%
VAGINAL BLEEDING	6	6.32%
HYPERTENSION	7	7.36%
DIABETES MELLITUS	3	3.15%
GESTATIONAL DIABETES	3	3.15%
PREECLAMPSIA	1	1.05%

Table 10 Distribution of study population based on risk factors related to mother

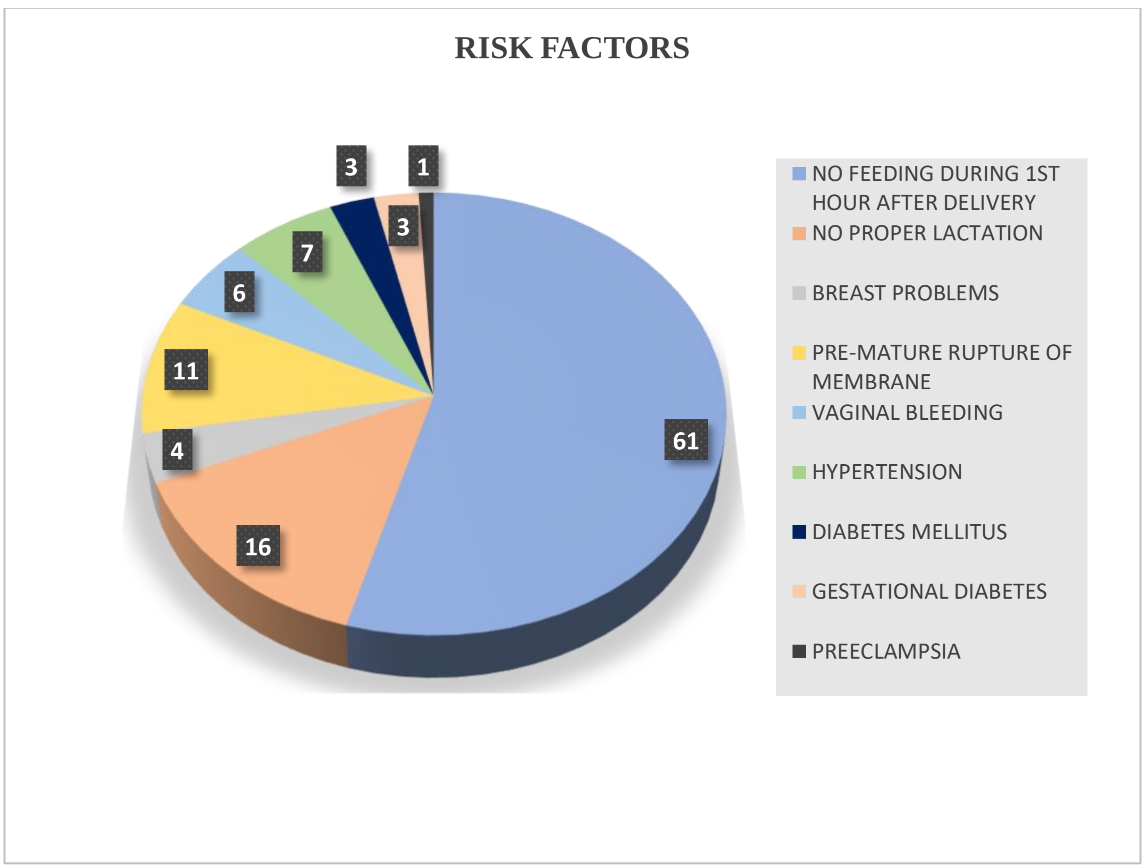


Figure 16 Distribution of study population based on risk factors related to mother

Considering the risk factors related to the mother, 64.21% of the mothers did not feed their babies for the 1st one hour after delivery, 16.84% had a problem of proper lactation, 4.21% of mothers have breast problems, 11.58% of mothers had premature rupture of membrane, 6.36% of them had vaginal bleeding, 7.36% of the mothers have hypertension, 3.15% of them with diabetes mellitus and the same percentage of mothers with gestational diabetes i.e., 3.15% and 1.05% of the subjects with preeclampsia.

6.11 DISTRIBUTION OF STUDY POPULATION BASED ON RISK FACTORS RELATED TO NEONATES

RISK FACTOR	NUMBER OF SUBJECTS		TOTAL
	FULLTERM (%)	PRETERM (%)	
PREVIOUS SIBLING WITH JAUNDICE	20(31.74%)	10(31.25%)	30(31.57%)
SEPSIS	4(6.34%)	0(0%)	4(4.21%)
DIFFICULTY IN FEEDING	10(15.87%)	6(18.75%)	16(16.84%)
MALE SEX	35(55.55%)	14(43.75%)	49(51.57%)
ABO INCOMPATIBILITY	8(12.69%)	5(15.62%)	13(13.68%)
RH IMCOMAPTIBILITY	4(6.34%)	2(6.25%)	6(6.31%)

Table 11 Distribution of study population based on risk factors related to neonates

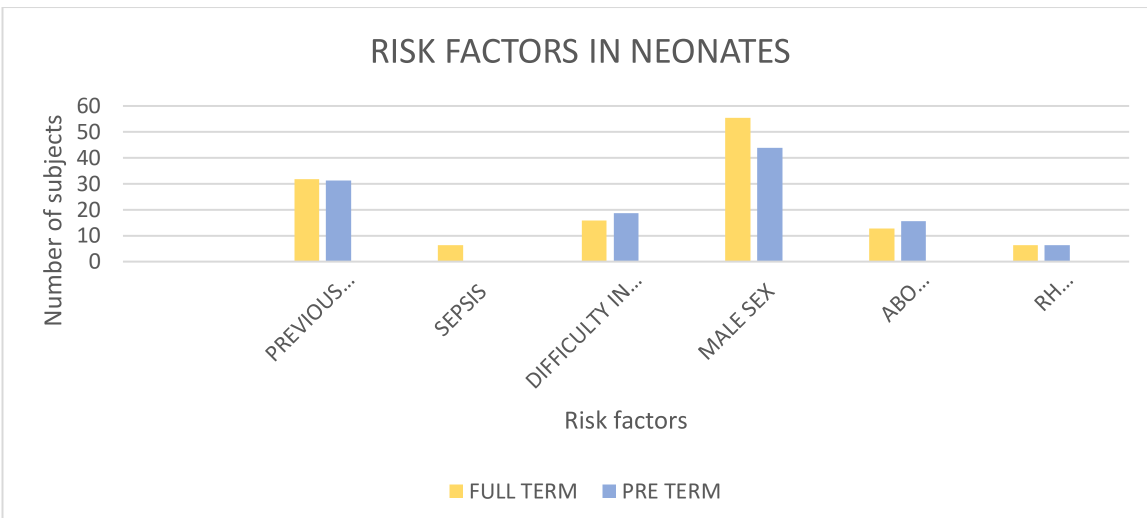


Figure 17 Distribution of study population based on risk factors related to neonates

Considering the risk factors related to neonates, 31.57% of the neonates have a history of previous sibling with jaundice, 4.21% of the neonates had sepsis, 16.84% of them had difficulty in feeding, 51.57% of the neonates are of male sex, 13.68% of them have ABO incompatibility and 6.31% of neonates have Rh incompatibility.

6.12 DISTRIBUTION OF STUDY POPULATION(NEONATES) BASED ON BILIRUBIN LEVELS ON DAY OF ADMISSION

TOTAL SERUM BILIRUBIN (mg/dl)	FULL-TERM (%)	PRE-TERM (%)	GRAND TOTAL (%)
10-11	3(4.76%)	3(9.38%)	6(6.32%)
11-12	5(7.94%)	3(9.38%)	8(8.42%)
12-13	18(28.57%)	11(34.38%)	29(30.53%)
13-14	18(28.57%)	8(25.00%)	26(13.68%)
14-15	9(14.29%)	4(12.50%)	13(13.68%)
15-16	7(11.11%)	2(6.25%)	9(9.47%)
16-17	3(4.76%)	1(3.13%)	4(4.21%)
GRAND TOTAL	**63**	**32**	**95**

Table 12 Distribution of study population(neonates) based on bilirubin levels on day of admission

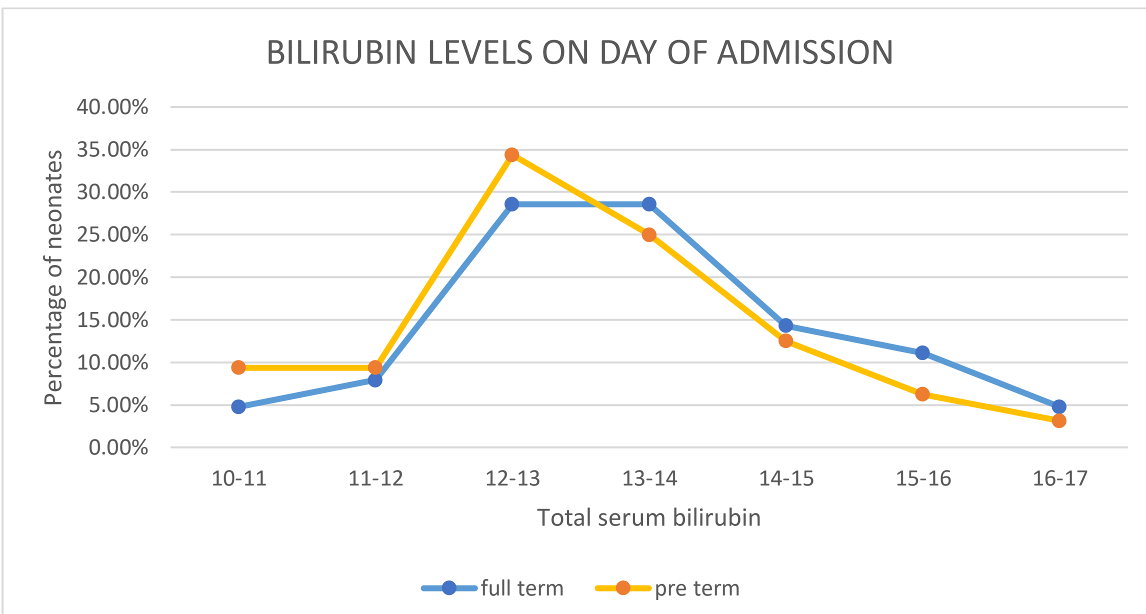


Figure 18 Distribution of study population(neonates) based on bilirubin levels on day of admission

During the study, majority of full-term neonates had bilirubin levels of 12-14mg/dl and pre-term neonates had bilirubin levels of 12-13mg/dl on day of admission. Least number of full-term neonates had bilirubin levels of 10-11mg/dl and 16-17mg/dl and pre-term neonates had bilirubin levels of 16-17mg/dl on day of admission.

6.13 DISTRIBUTION OF STUDY POPULATION(NEONATES) BASED ON BILIRUBIN LEVELS ON DAY OF DISCHARGE

TOTAL SERUM BILIRUBIN (mg/dl)	FULL TERM (%)	PRE-TERM (%)	GRAND TOTAL (%)
3-4	1(1.59%)		1(1.05%)
4-5	2(3.17%)		2(2.11%)
5-6	6(9.52%)	5(15.63%)	11(11.58%)
6-7	10(15.87%)	7(21.88%)	17(17.89%)
7-8	14(22.22%)	4(12.50%)	18(18.95%)
8-9	12(19.05%)	2(6.25%)	14(14.74%)
9-10	9(14.29%)	8(25.00%)	17(17.89%)
10-11	4(6.35%)	4(12.50%)	8(8.42%)
11-12	3(4.76%)	2(6.25%)	5(5.26%)
12-13	2(3.17%)		2(2.11%)
GRAND TOTAL	**63**	**32**	**95**

Table 13 Distribution of study population(neonates) based on bilirubin levels on day of discharge

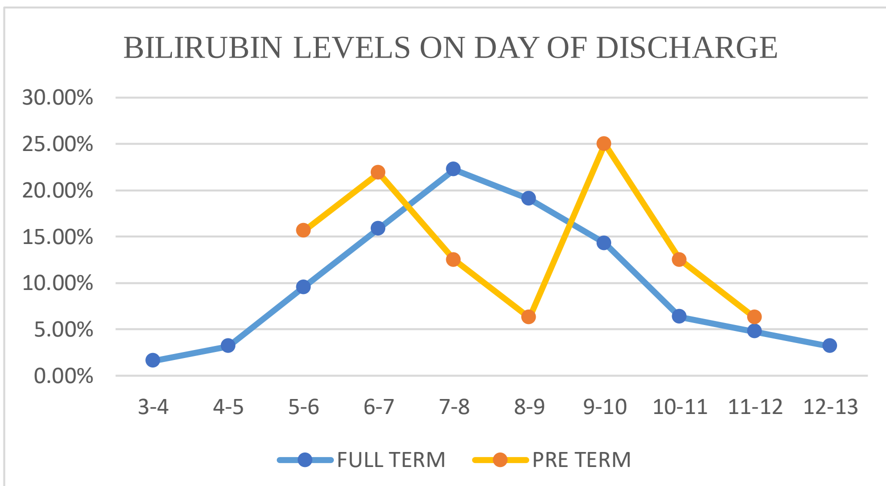


Figure 19 Distribution of study population(neonates) based on bilirubin levels on day of discharge

During the study, majority of full-term neonates had bilirubin levels of 7-8mg/dl and pre-term neonates had bilirubin levels of 9-10mg/dl on day of discharge. Least number of full-term neonates had bilirubin levels of 3-4mg/dl and pre-term neonates had bilirubin levels of 3-4 mg/dl and 11-12mg/dl on day of discharge.

6.14 KNOWLEDGE OF MOTHERS ON NEONATAL JAUNDICE

QUESTION	NUMBER OF SUBJECTS	PERCENTAGE
Does mother know anything about neonatal jaundice?	68	71.58%
Does mother know any sign or symptom of jaundice?	68	71.58%
Do mothers think whether neonatal jaundice is normal?	60	63.16%
Does the mother know seriousness of neonatal jaundice?	20	21.05%
Does mother know any treatment option for neonatal jaundice?	55	57.89%

Table 14 Knowledge of mothers on neonatal jaundice

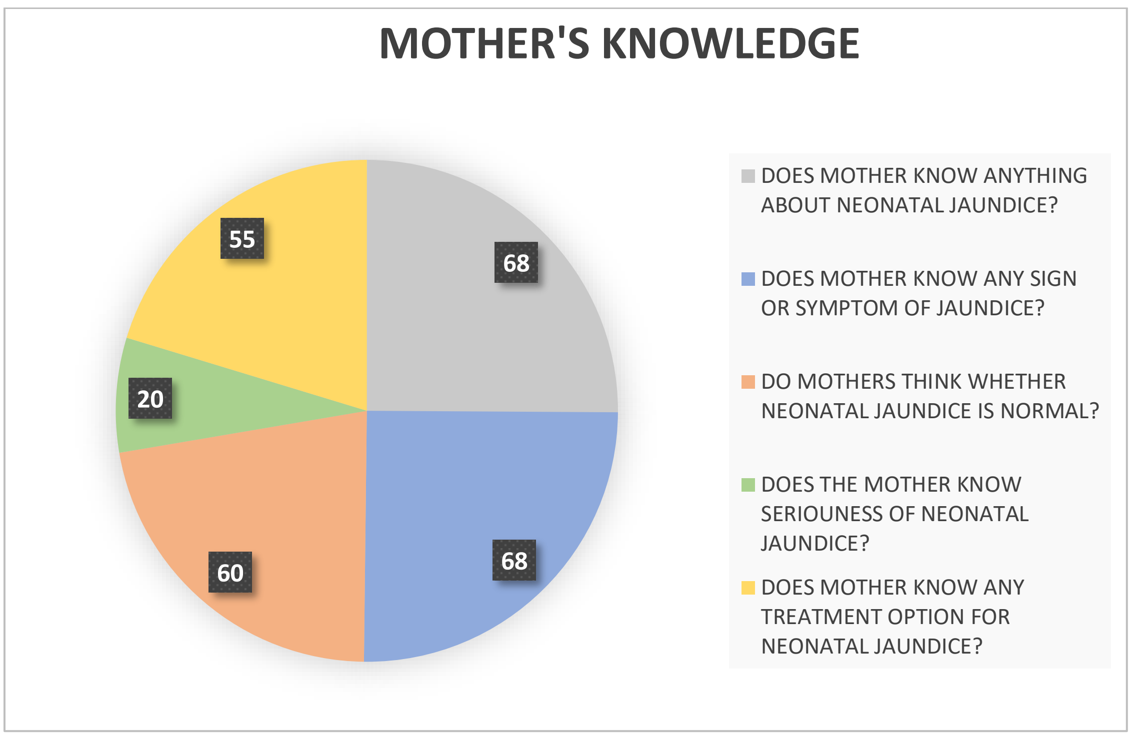


Figure 20 Knowledge of mothers on neonatal jaundice

In the total study population 71.58% of mothers have the knowledge of neonatal jaundice, 57.89% have the knowledge of treatment for neonatal jaundice and only 21.05% know the seriousness of jaundice.

6.15 KNOWLEDGE OF MOTHERS ABOUT SIGNS AND SYMPTOMS OF NEONATAL JAUNDICE

SIGNS/SYMPTOMS	NUMBER OF SUBJECTS	PERCENTAGE
YELLOW EYES	36	37.89%
YELLOW SKIN	19	20.00%
YELLOW SKIN AND EYES	10	10.53%
YELLOW FAECES	1	1.05%
YELLOW URINE	5	5.26%
FEEDING DIFFICULTY	1	1.05%
VOMITING AND DIARRHOEA	1	1.05%
NO FAECES AND URINE	1	1.05%

Table 15 Knowledge of mothers about signs and symptoms of neonatal jaundice

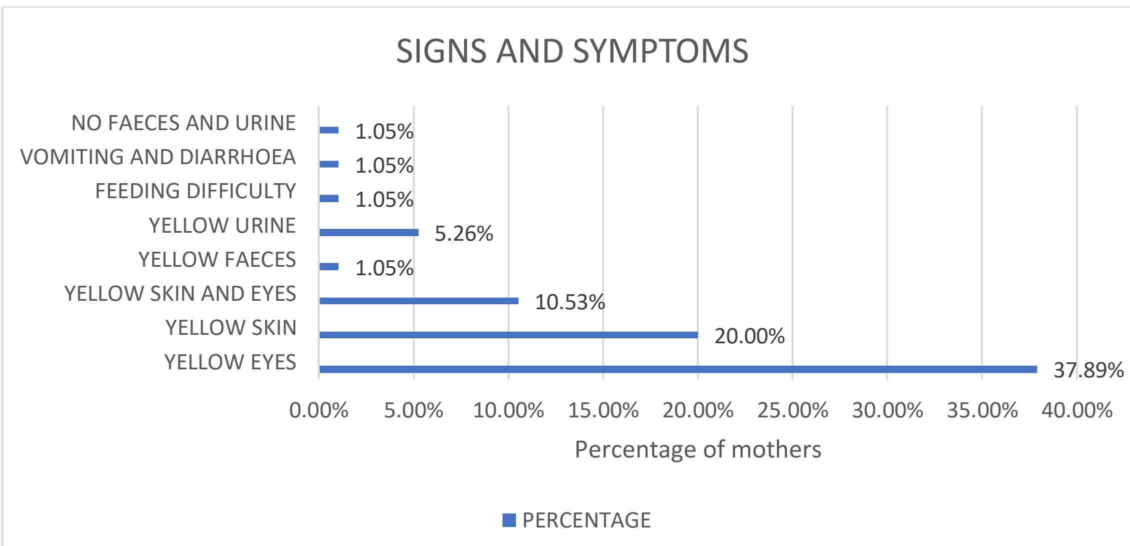


Figure 2 Knowledge of mothers about signs and symptoms of neonatal jaundice

During the study the highest number of mothers answered yellow eyes (37.89%) as signs and symptoms, 20% answered as yellow skin and 10.53% answered both yellow skin and eyes, 5.26% of the mothers knew yellow urine as symptom,1.05% of the mothers answered as yellow feces, difficulty in feeding, vomiting and diarrhea, no feces and urine as signs and symptoms of jaundice.

6.16 KNOWLEDGE OF MOTHERS ON TREATMENT OF NEONATAL JAUNDICE

TREATMENT	NUMBER OF PATIENTS	PERCENTAGE
SUNLIGHT	42	44.21%
PHOTOTHERAPY	17	17.89%
WARM CARE	1	1.05%
BREAST MILK	1	1.05%

Table 16 Knowledge of mothers on treatment of neonatal jaundice

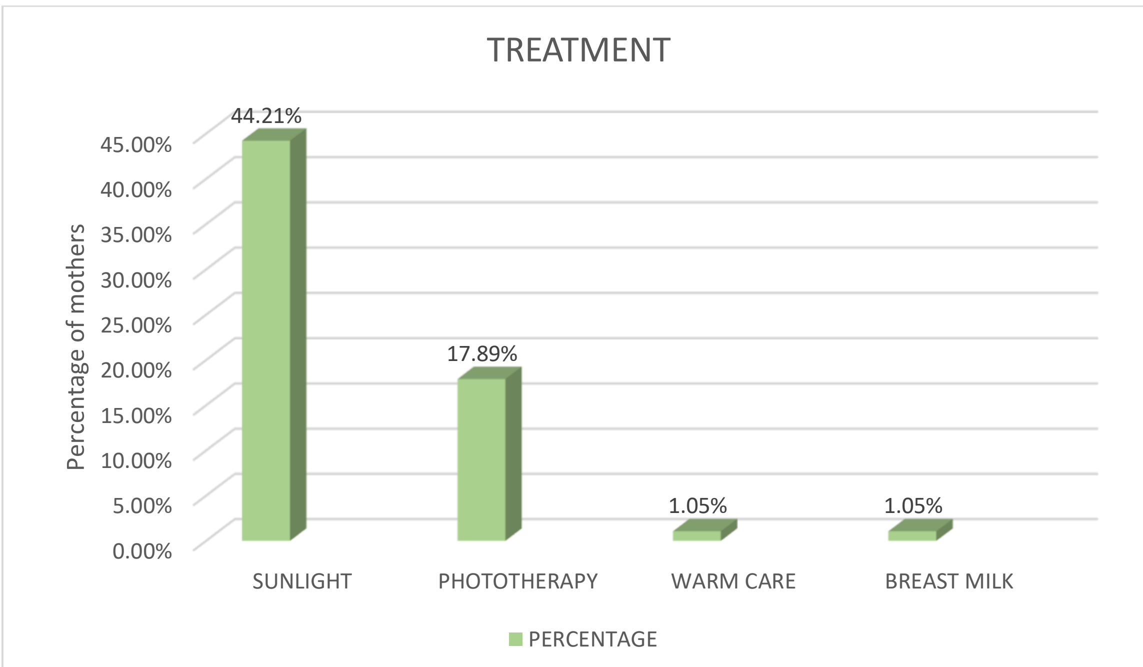


Figure 22 Knowledge of mothers on treatment of neonatal jaundice

In the study population, majority of mothers knew the treatment for jaundice is sunlight (44.21%), 17.89% of mothers have knowledge of phototherapy as the treatment for neonatal jaundice and 1.05% of mothers answered as warm care and breast milk.

DISCUSSION

7.DISCUSSION

DISTRIBUTION OF STUDY POPULATION (MOTHERS AND NEONATES) BASED ON DEMOGRAPHIC DETAILS

The most numbers of study subjects are in 25-29 age groups with 45.26% of the total study population. The least number of study subjects are in 35-39 age groups with 8.42% of the total study population. The mean age of the mothers was found to be 27.610± 4.354.

In the total number of study subjects, 49(51.58%) are male neonates and 46(48.42%) are female neonates.

In the total study population, majority of subjects are from urban area (83.16%) and subjects belonging to rural area (16.84%).

In the total study population, the marital life of 82 subjects is non-consanguineous with 86.32% and 13 subjects is consanguineous with 13.68%.

During the study period, it was found that 46 study subjects have completed their education till graduation, 9 subjects till post-graduation, 13 completed till intermediate, 2 of them completed their diploma and 25 subjects completed their schooling.

DISTRIBUTION OF STUDY POPULATION BASED ON BIRTH WEIGHT AND PRE-TERM OR FULL TERM

In the total study population, the majority of full-term neonates had birth weight of 3-3.2 kgs and least number of study subjects were 2.2-2.4 kgs. In pre term neonates, majority of study subjects had birth weight of 2.8- 3 kgs and least number of them had birthweight of 2-2.2 kgs. The mean birth weight of full-term neonates is 2.855± 0.396 and in pre term is 2.857± 0.398.

In a systemic review performed by Hassan Boskabadi, et al. have found that prematurity was associated with neonatal jaundice.[16]

DISTRIBUTION OF STUDY POPULATION BASED ON TYPE OF DELIVERY

The greatest number of study subjects had undergone cesarean type of delivery with 93.68% and the rest of the subjects had undergone normal delivery with 6.32%.

In retrospective cross-sectional design, Murekatete C, et al. have found that cesarean type of delivery was one of the risk factors for neonatal jaundice.[11]

DISTRIBUTION OF STUDY POPULATION BASED ON INTERPREGANANCY GAP

During the study, majority of the study subjects had an interpregnancy gap of 0-1 year with 35.79% and lowest number of subjects had 7-9 years gap with 1.05%. The mean interpregnancy gap is 1.920± 1.992.

DISTRIBUTION OF STUDY POPULATION BASED ON DURATION OF JAUNDICE

In the study period, majority of the subjects had highest duration of jaundice for 2 days with a percentage of 41.05%. The mean duration of jaundice in neonates is 2.273± 0.961.

DISTRIBUTION OF STUDY POPULATION BASED ON RISK FACTORS RELATED TO MOTHER

The most common risk factor related to mothers was found to be no feeding during 1st hour after delivery (64.21%) and the least common risk factor was found to be preeclampsia (1.05%).

Similar study was performed by Hassan Boskabadi, et al. The study concluded with list of maternal risk factors includes pregnancy problems(prematurity, premature rupture of membrane, negative maternal Rh), maternal diseases(hypertension, diabetes, preeclampsia, gestational infections), labor problems(type of delivery, cephalohematoma, place of delivery, delayed labor, delivery trauma, complications), breastfeeding problems(breast problems, reduced feeding, inappropriate breastfeeding technique) and other factors like maternal age, first delivery, abortion history.[16]

DISTRIBUTION OF STUDY POPULATION BASED ON RISK FACTORS RELATED TO NEONATES

The most common risk factor in both full term and pre term neonates was male sex with percentages of 55.55% and 43.75% respectively. The least common risk factor in full term neonates was sepsis and Rh incompatibility (6.34%) and in pre term neonates was Rh incompatibility (6.25%).

Similar studies were performed by Murekatete C, et al. This study found that Gestational age, ABO and other blood group incompatibilities, Birth weight, neonatal gender, infections, prematurity and C-section were associated with neonatal jaundice.[11]

In another study performed by Asmamaw demis bizureh, et al. found out prolonged labour, low birth weight, sepsis, birth asphyxia, hypothermia, sex were risk factors of Neonatal jaundice.[23]

Study performed by Stephanee campbell wagemann, patricia mena nannig found out male sex, prematurity, excessive weight loss and blood group incompatibility were the risk factors for severe hyperbilirubinemia.[25]

DISTRIBUTION OF STUDY POPULATION(NEONATES) BASED ON BILIRUBIN LEVELS ON DAY OF ADMISSION

During the study, majority of full-term neonates had bilirubin levels of 12-14mg/dl and pre-term neonates had bilirubin levels of 12-13mg/dl on day of admission. Least number of full-term neonates had bilirubin levels of 10-11mg/dl and 16-17mg/dl and pre-term neonates had bilirubin levels of 16-17mg/dl on day of admission.

The average total serum bilirubin on day of admission was found to be 13.189± 1.407.

DISTRIBUTION OF STUDY POPULATION(NEONATES) BASED ON BILIRUBIN LEVELS ON DAY OF DISCHARGE

During the study, majority of full-term neonates had bilirubin levels of 7-8mg/dl and pre-term neonates had bilirubin levels of 9-10mg/dl on day of discharge. Least number of full-term neonates had bilirubin levels of 3-4mg/dl and pre-term neonates had bilirubin levels of 3-4 mg/dl and 11-12mg/dl on day of discharge. The average total serum bilirubin on day of discharge was found to be 8.015± 1.915.

KNOWLEDGE OF MOTHERS ON NEONATAL JAUNDICE

In the total study population 71.58% of mothers have the knowledge of neonatal jaundice, 57.89% have the knowledge of treatment for neonatal jaundice and only 21.05% know the seriousness of jaundice.

Similar studies were performed by Goodman et.al, Asmamaw Demis et.al, and Egube BA et.al, the studies found that the awareness of neonatal jaundice among mothers is 75.4%, 39.2% and 85.9% respectively.[12][14][19]

KNOWLEDGE OF MOTHERS ABOUT SIGNS AND SYMPTOMS OF NEONATAL JAUNDICE

During the study the highest number of mothers answered yellow eyes (37.89%) as signs and symptoms of neonatal jaundice.

In a cross-sectional study by Amegan-Aho KH, et al. 72.6%of mothers knew at least any one sign or symptom.[22]

KNOWLEDGE OF MOTHERS ON TREATMENT OF NEONATAL JAUNDICE

In the study population, majority of mothers knew the treatment for jaundice is sunlight (44.21%).

In a study performed by Amegan-Aho KH, et al. 52% of mothers know any one of treatment options.[22]

ASSOCIATION OF MOTHER'S KNOWLEDGE WITH DIFFERENT MATERNAL FACTORS

We analysed mother's knowledge with different maternal factors like age (p value = 0.221782), locality (p value= 0.78322), primiparity (p value =0.000167) and education (p value= 0.000736).

We got significant association of mother's knowledge with primiparity and education.

ASSOCIATION OF TOTAL SERUM BILIRUBIN WITH MATERNAL RISK FACATORS FOR NEONATAL JAUNDICE

We analyzed association of total serum bilirubin with difficulty in feeding (p value =0.079864), no feeding during 1st hour after delivery (p value=0.906782), PROM (p value = 0.136583), marital life (p value=0.806963) and gestational diabetes (p value = 0.000131).

We got significant association between gestational diabetes in mother and total serum bilirubin in neonates.

ASSOCIATION OF TOTAL SERUM BILIRUBIN WITH NEONATAL RISK FACATORS FOR NEONATAL JAUNDICE

We analyzed association of total serum bilirubin with sex (p value = 0.570795), Rh incompatibility (p value=0.368256), ABO incompatibility (p value=0.456893), previous sibling with jaundice (p value = 0.061783), term of pregnancy (p value= 0.181818) and sepsis (p value= 0.116545).

CONCLUSION

8.CONCLUSION

Neonatal jaundice is yellowish discoloration of skin, conjunctiva and sclera due to elevated serum or plasma bilirubin in the newborn period. In our study, the most common risk factor among mothers is no feeding during 1^{st} hour after delivery and among neonates is male sex. Mothers had awareness about neonatal jaundice but knowledge regarding complications and treatment was low. The awareness about neonatal jaundice should be increased among mothers to identify, for early diagnosis and treatment. During our study, we helped mothers understand better about Neonatal jaundice.

LIMITATIONS

9.LIMITATIONS

- Larger sample size is needed to draw proper conclusions which could have been possible with larger study period.
- We got less sample size which is needed to give definite conclusions due to third wave of COVID-19.

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ANNEXURE I

NEONATAL JAUNDICE

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1

DEFINITION:

Neonatal jaundice is yellowish discoloration of skin and white portion of eyes due to elevated bilirubin in blood in the new born

CAUSES:

Either pathologic or physiologic

Physiologic due to –

- Differences in metabolism of bilirubin in neonatal period leading to increased bilirubin level

Pathological due to –

- Breakdown of blood which is immune mediated such as ABO & Rh incompatibility
- Cephalohematoma- accumulation of blood in between skull and brain
- Extensive bruising

2

SYMPTOMS:

Yellowing

- In the whites of their eyes
- On the soles of their feet
- Inside their mouth
- On the palms of their hands

A new born may also

- Be sleepy
- Not want to feed or not feed as well as usual
- Have dark, yellow urine
- Have pale stools

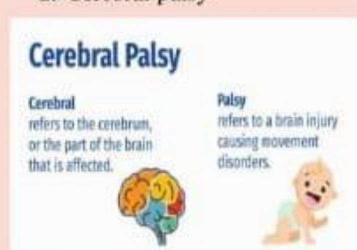
3

COMPLICATIONS:

1. Kemicterus-

It is an extremely rare bilirubin induced brain damage, which occurs in newborns suffering from severe jaundice.

2. Cerebral palsy-



3. Deafness

4

TREATMENT:

1. Phototherapy-

The eyes and genetilia of new born are covered and the body surface area is exposed to light by which the bilirubin is excreted in urine.

2. Exchange transfusion-

It is the blood transfusion where the bilirubin is removed from the circulation.

5

3. IV Immunoglobulin-

It reduces the need of exchange transfusion used for reducing the break down of blood and consequent increase in bilirubin in Rh immunisation and ABO incompatibility.

4. IV hydration and other agents like phenobarbitone can be given.

PREVENTION:

- Promote and support breast feeding
- Advise a frequency of 8-10 feedings per day
- If mother can't breast feed Formula milk for term infants should be 1 to 2 ounce every 2-3 hrs in the 1st week
- Routine supplements with warm water or dextrose water will not help prevent increase bilirubin.

6

నవజాత కామెర్లు

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1

నిర్వచనం:

నియోనాటల్ కామెర్లు అనేది నవజాత శిశువులో రక్తంలో పెరిగిన బిలిరుబిన్ కారణంగా చర్మం మరియు కంటి తెలుపు పసుపు రంగులోకి మారుతుంది

కారణాలు:

- నవజాత కాలంలో బిలిరుబిన్ యొక్క జీవక్రియలో తేడాలు బిలిరుబిన్ స్థాయిని పెంచుతాయి
- ABO & Rh అననుకూలత వంటి రోగనిరోధక మధ్యవర్తిత్వం కలిగిన రక్తం విచ్ఛిన్నం కావడం
- సెఫలోహెమటోమా - పుర్రె మరియు మెదడు మధ్య రక్తం చేరడం
- విస్తృతమైన గాయాలు

2

లక్షణాలు:

పసుపురంగు

- వారి కళ్ళలోని తెల్లటి రంగులో
- వారి పాదాల మీద
- వారి నోటి లోపల
- వారి అరచేతులపై

-నిద్ర మత్తు

-తిండికి ఇష్టపడరు

-మూత్రం ముదురు పసుపు రంగులో ఉంటుంది

-లేత రంగు మలం కలిగి ఉంటాయి

3

సంక్లిష్టతలు:

1. Kernicterus-

ఇది చాలా అరుదైన బిలిరుబిన్ ప్రేరిత మెదడు నష్టం, ఇది తీవ్రమైన కామెర్లుతో బాధపడుతున్న నవజాత శిశువులలో సంభవిస్తుంది.

2. సెరిబ్రల్ పాల్సీ-

మెదడు ప్రభావితమై కదలిక రుగ్మతలకు కారణమవుతుంది

3. చెవుడు

చికిత్స:

1. ఫోటో థెరపీ-

నవజాత శిశువు యొక్క కళ్ళు మరియు జననేంద్రియాలు కప్పబడి ఉంటాయి మరియు

4

శరీర ఉపరితల వైశాల్యం కాంతికి గురవుతుంది, దీని ద్వారా బిలిరుబిన్ మూత్రంలో విసర్జించబడుతుంది.

2. రక్త మార్పిడి -

ఇక్కడ రక్త ప్రసరణ నుండి బిలిరుబిన్ తొలగించబడుతుంది.

3. IV ఇమ్యునోగ్లోబులిన్-

ఇది రక్తం యొక్క విచ్ఛిన్నతను తగ్గించడానికి ఉపయోగించే రక్త మార్పిడి అవసరాన్ని తగ్గిస్తుంది మరియు Rh రోగనిరోధక

5

మరియు ABO అనుకూలతలో బిలిరుబిన్ పెరుగుదలను తగ్గిస్తుంది.

4. IV హైడ్రేషన్ మరియు ఫెనోబార్బిటోన్ వంటి ఇతర ఏజెంట్లు ఇవ్వవచ్చు

నివారణ:

- తల్లి పాలివ్వడాన్ని ప్రోత్సహించండి మరియు మద్దతు ఇవ్వండి
- రోజుకు 8-10 ఫీడింగ్ ఫ్రీక్వెన్సీని సూచించండి
- తల్లి పాలివ్వలేని పక్షంలో 1వ వారంలో ప్రతి 2-3 గంటలకు 1 నుండి 2 ఔన్సుల వరకు శిశువులకు ఫార్ములా పాలు ఇవ్వాలి.
- గోరు వెచ్చని నీరు లేదా చక్కెర నీటితో సాధారణ సప్లిమెంట్లు బిలిరుబిన్ పెరుగుదలను నిరోధించడంలో సహాయపడవు.

6

नवजात पीलिया

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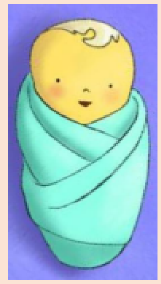
परिभाषा:



नवजात पीलिया नवजात के रक्त में बिलीरुबिन बढ़ने के कारण त्वचा और आंखों का सफेद पीला रंग हो जाएगा।

कारण:

- नवजात शिशु में बिलीरुबिन के चयापचय में अंतर से बिलीरुबिन का स्तर बढ़ जाता है
- ABO और Rh असंगतता जैसे प्रतिरक्षा मध्यस्थ रक्त के थक्के
- सेफलोहेमेटोमा - खोपड़ी और मस्तिष्क के बीच रक्त का संचय
- व्यापक चोटें

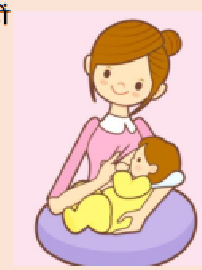
2

विशेषताएं:

पीला रंग

- उनकी आंखों की सफेदी में
- अपने पैरों पर
- उनके मुंह के अंदर
- उनके हाथों की हथेलियों पर

- नींद का नशा
- खाना पसंद नहीं
- मूत्र काला पीले रंग में होगा
- हल्के रंग का मल पास होना

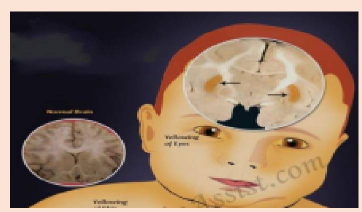


3

जटिलताएं:

1. कर्निकटेरस-

यह एक बहुत ही दुर्लभ बिलीरुबिन-प्रेरित मस्तिष्क क्षति है जो गंभीर पीलिया वाले नवजात शिशुओं में होती है।.



2. सेरेब्रल पाल्सी-

मस्तिष्क प्रभावित होता है और गति विकारों का कारण बनता है

3. बहरा

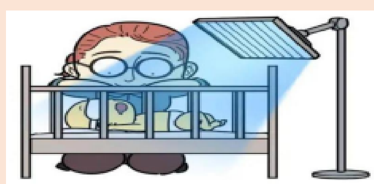
इलाज:

1. फोटो थेरेपी-

नवजात शिशु की आंखें और जननांग ढके होते हैं और शरीर की सतह का क्षेत्र प्रकाश के संपर्क में आता है,

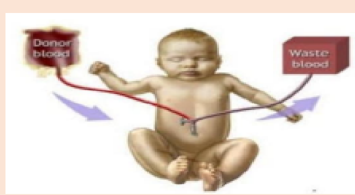
4

जिसके माध्यम से बिलीरुबिन मूत्र में उत्सर्जित होता है



2. रक्त आधान -

यहां खून से बिलीरुबिन को हटा दिया जाता है।



3. IV इम्युनोग्लोबुलिन-

यह रक्त के टूटने और आरएच प्रतिरक्षण को कम करने के लिए उपयोग किए जाने वाले रक्त आधान की आवश्यकता को कम करता है और एबीओ संगतता में बिलीरुबिन वृद्धि को कम करता है।

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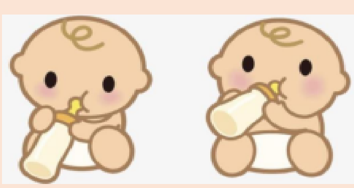
कम करता है और एबीओ संगतता में बिलीरुबिन वृद्धि को कम करता है।

4. IV हाइड्रेशन और अन्य एजेंट जैसे फेनोबार्बिटोन दिया जा सकता है

निवारण:

- स्तनपान को प्रोत्साहित और समर्थन करें
- प्रति दिन 8-10 फीडिंग की आवृत्ति निर्दिष्ट करें
- अगर मां स्तनपान नहीं करा पा रही है तो शिशुओं को पहले सप्ताह के दौरान हर 2-3 घंटे में 1 से 2 औंस फॉर्मूला दूध पिलाना चाहिए।
- गुनगुने पानी या चीनी के पानी के साथ नियमित सप्लीमेंट बिलीरुबिन के विकास को रोकने में मदद नहीं करते हैं।



6

ANNEXURE II

A PROSPECTIVE OBSERVATIONAL STUDY ON RISK ASSESMENT AND MOTHER'S KNOWLEDGE ON NEONATAL JAUNDICE

MOTHER'S DETAILS:

Name:	Age:	Sex:
Department:	Date of admission:	Date of discharge:
IP/OP number:	Residency: rural ☐ urban ☐	

Past medical history: Hypertension ☐ Diabetes mellitus ☐Asthma ☐ Epilepsy☐ Thyroid disorders ☐
Respiratory Disorders☐ Kidney Disorders☐ Liver Disorders☐ Others☐
____

Past medication history: ____

Blood group:________ If Rh negative, then did mother take anti-D: Yes ☐ No ☐
Mode of conception: ________________________
Number of pregnancies: __________ Number of live children: _______ Interpregnancy gap: ______
Marital life: Consanguineous ☐ Non- Consanguineous ☐
Education qualification: ____________________

NEONATE DETAILS:

Name:	Age:	Sex:
Department:	Date of admission:	Date of discharge:
IP/OP number:	Admission weight:	Birth weight:

Percentage of weight loss: ___________
Blood group: ___________ Is there any incompatibilities with mother? Yes ☐ No ☐
Full term ☐ pre term ☐
Duration of jaundice: ______________ How many days after birth was jaundice seen? _________
Difference/difficulty in feeding: Yes ☐ No ☐
Treatment given: ______________

Bilirubin levels:

Days										

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MATERNAL RISK FACTOR ASSESMENT FOR NEONATAL JAUNDICE

1. Age: ______
2. Weight: ________
3. Pre term ☐ full term ☐ If pre term, by how many weeks? ____________________
4. Blood group of mother: ___ Blood group of baby:___ Is there any incompatibilities? Yes ☐ No ☐
5. Preeclampsia: Yes ☐ No☐
6. Hypertension: Yes ☐ No ☐
7. Diabetes mellitus: Yes ☐ No ☐ if yes, then gestational diabetes ☐ or known case of diabetes ☐
8. Vaginal bleeding: Yes ☐ No☐ If yes, then from how many days? ________
9. Type of delivery: Normal vaginal delivery ☐ Cesarean section ☐
10. Place of delivery: ____________________
11. Initiation of breast feeding during 1st hour of life: Yes☐ No ☐
12. Premature rupture of membrane (PROM): Yes ☐ No ☐ If yes, what is the duration? ________
13. Proper lactation: Yes ☐ No ☐
14. Breast problems: Yes ☐ No ☐ If yes, what? ________________

NEONATAL RISK FACTOR ASSESMENT FOR NEONATAL JAUNDICE

1. Birth weight: ___________
2. Sex: ________
3. Gestational age: _________
4. Cephalohematoma: Yes ☐ No☐
5. Difficulty in feeding: Yes ☐ No ☐
6. Blood group: _______ Is there any incompatibility? ABO ☐ Rh☐ No ☐
7. Gastrointestinal obstruction: Yes ☐ No☐
8. Sepsis: Yes ☐ No ☐
9. Hypothyroidism: Yes ☐ No☐
10. Enzyme deficiency: Yes ☐ No ☐ If yes, then what deficiency? ________________________
11. Autoimmune/hemolytic anaemia: Yes ☐ No ☐
12. Previous sibling with significant jaundice: Yes ☐ No ☐

MOTHER'S KNOWLEDGE ABOUT JAUNDICE

1. Does mother know anything about jaundice? Yes ☐ No ☐ If yes, then what? ____________
2. Do you know any signs and symptoms? Yes ☐ No ☐ If yes, then what? ____________
3. Do you think jaundice is normal? Yes ☐ No ☐
4. Do you know the seriousness of jaundice? Yes ☐ No ☐
5. Do you know anything about treatment? Yes ☐ No ☐ If yes, then what? ________________

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ANNEXURE 3

PATIENT CONSENT FORM (ANNEXURE-I) [i]

A PROSPECTIVE OBSERVATIONAL STUDY ON RISK ASSESSMENT AND MOTHER'S KNOWLEDGE ON NEONATAL JAUNDICE

Name of the patient:

Investigator/s name: Gargi Mondal, Kunche Sandhya, Manchala Ravi Teja, Rajanala Rathnakar

Name of the institution: Marri Laxman Reddy Institute of Pharmacy

Name of the research coordinators: Dr. Gabriela Keerthana Gondhi, Pharm. D, MBA - HSSM

For the patient:

I, ______________________________ exercising my free power of choice, hereby give my consent to be included as a patient in the clinical study, " A PROSPECTIVE OBSERVATIONAL STUDY ON RISK ASSESSMENT AND MOTHER'S KNOWLEDGE ON NEONATAL JAUNDICE", ESI HOSPITAL-SANATHNAGAR located at Nacharam.

I agree to the following:

1. I am over 18 years of age.
2. I understand that I will not be given any new medication for participation in this study.
3. I have been informed to my satisfaction by the attending investigator about the purpose of this study and study procedures.
4. I have been given a full explanation by supervising investigator of the nature, purpose, likely duration of the study and about what I will be expected to do. I have understood the information given to me.
5. I have been given the opportunity to question the attending investigator on all aspects of the study, and I have understood the advice and information as a result.
6. I am also aware of my right to opt out of the study at any time without giving any reason for doing so.

I have confirm that I have explained the nature, purpose of the above study to the fullest.

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Signature of the patient with date   **Signature of the caretaker with date**

Signature of the investigators with date

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PATIENT CONSENT FORM (ANNEXURE-I) [ii]

రోగి <u>అంగీకారపత్రం</u>

రోగి పేరు:

పరిశోధకుడి/ల పేరు: గార్గి మొండల్, కుంచె సంధ్య, మందాల రవితేజ, రాజనాల రత్నాకర్

పరిశోధన సమన్వయకర్త: Dr. గాబ్రియేలా కీర్తన గెంది, ఫార్మ్.డి , MBA-

HSSM

<u>రోగి కొరకు:</u>

నేను ____________________, నా స్వేచ్ఛ హక్కులు వినియోగించి ఇందుమూలముగా తెలియజేయునది ఏమిటంటే వీరు చేసేతున్న " నవజాత కామెర్లుపై ప్రమాద అంచనా మరియు తల్లికి సంబంధించిన అవగాహనపై భావి పరిశీలనా అధ్యయనం", నేను నా పూర్తి అనుమతి తెలుపుతున్నాను.

<u>నేను ఈ కింది వాటిని అంగీకరిస్తున్నాను :</u>

1. నా వయసు, 18 సంవత్సరములకు పైన.
2. ఈ అధ్యయనంలో పాల్గొనడాన్కి నాకు కొత్త మందులు ఇవ్వబడవన్ నేను అర్థం చేసుకున్నాను.
3. ఈ అధ్యయనం మరియు అధ్యయన విధానాల ఉద్దేశ్యం గురించి హాజరైన పరిశోధకుడి ద్వారా నా సంతృప్తికి తెలియజేయబడింది.
4. ఈ అధ్యయనం యొక్క స్వభావం, ఉద్దేశ్యం, వ్యవధి మరియు నేను ఏమి చేయబోతున్నానో దాన్ గురించి పరిశోధకుడి ద్వారా నాకు పూర్తి వివరణ ఇవ్వబడింది. నాకు ఇచ్చిన సమాచారం నాకు అర్థమైంది.
5. నాకు ఈ అధ్యయనం లోన్ అన్ని అంశాల గురించి పరిశోధకుడిన్ ప్రశ్నించే పూర్తి హక్కు ఇవ్వబడింది మరియు నేను ఫలితంగా సలహా
6. మరియు సమాచారాన్ని అర్థం చేసుకున్నాను.
7. ఈ అధ్యయనం నుండి నేను ఏ సమయమునైన, ఎందు మూలముగానైనా వీరికి ఏ వివరణ ఇవ్వకుండా వైదొలగవచ్చాన్ నాకు తెలుసు. ఈ అధ్యయనం, దాన్ వ్యవధి, ఉద్దేశ్యం, ప్రయోజనం గురించి నాకు **తెలియజేయబడింది.**

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రోగి సంతకం / తేదీ *** సంరక్షకుని యొక్క సంతకం / తేదీ**

పరిశోధకుల సంతకం / తేదీ

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